The Heston Model

From Semi-Closed-Form Pricing to Calibration

Fourier inversion, Monte Carlo simulation, variance reduction and identifiability

Zakaryae Asserrar

Master 2 Probabilités et Finance
(ex-DEA El Karoui)
Sorbonne Université – École polytechnique



Abstract

We study the Heston model from end to end. We derive the characteristic function of the log-price and prove the Gil-Pelaez and Lewis formulas, which give a reference price accurate to 10^{-8}. We then compare the Euler and QE simulation schemes, in particular when the Feller condition is violated, and we build a variance reduction that divides the variance by more than 40 at the money. Finally, we calibrate the model on a synthetic market and then on AAPL options, and we show why a good fit does not guarantee identifiable parameters.

Contents

1 Introduction and foundations of the Heston model **3**
- 1.1 Motivation: going beyond the Black-Scholes framework . . . 3
- 1.2 Mathematical formulation of the model . . . 3
- 1.3 Financial interpretation of the parameters . . . 3
- 1.4 Properties of the variance process and the Feller condition . . . 4

2 Decomposition of the call price and change of numéraire **4**

3 Characteristic function of the log-price **5**
- 3.1 Why use the characteristic function? . . . 5
- 3.2 Dynamics of the log-price and intuition for the exponential-affine form . . . 5
- 3.3 Feynman-Kac equation and exponential-affine form . . . 5
- 3.4 Explicit solution . . . 6
- 3.5 Characteristic functions under $\mathbb{Q}$ and under $\widetilde{\mathbb{Q}}$ . . . 6

4 Gil-Pelaez inversion formula **7**
- 4.1 Statement and proof . . . 7
- 4.2 Application to the Heston model . . . 8

5 Lewis formula **8**
- 5.1 Motivation: the limitations of the Gil-Pelaez formula . . . 8
- 5.2 Statement and proof . . . 8
- 5.3 Comparison with Gil-Pelaez . . . 9

6 Numerical validation **10**
- 6.1 Numerical setting . . . 10
- 6.2 Checks on the characteristic function . . . 10
- 6.3 Consistency between the Gil-Pelaez and Lewis formulas . . . 10
- 6.4 Black-Scholes limit as $\xi \to 0$ . . . 10
- 6.5 Summary . . . 11

7 Monte Carlo simulation **12**
- 7.1 Why simulate when a semi-closed-form formula is available? . . . 12
- 7.2 Principle of the Monte Carlo estimator . . . 12
- 7.3 The Euler scheme and the positivity problem . . . 13
- 7.4 The exact law of the CIR process . . . 13
  - 7.4.1 Conditional moments . . . 13
  - 7.4.2 Laplace transform and noncentral chi-squared distribution . . . 14
  - 7.4.3 The parameter d and the Feller condition . . . 14
- 7.5 Why avoid the exact law in the simulation? . . . 15
- 7.6 Andersen's QE scheme for the variance . . . 15
  - 7.6.1 Quadratic branch (small ψ) . . . 16
  - 7.6.2 Exponential branch (large ψ) . . . 16
  - 7.6.3 Choice of branch and properties . . . 17
- 7.7 Discretisation of the log-price . . . 17
- 7.8 Martingale correction . . . 18
- 7.9 Numerical results of standard Monte Carlo . . . 18
  - 7.9.1 Test 1: law of the variance over one step . . . 18
  - 7.9.2 Test 2: discretisation bias . . . 19
  - 7.9.3 Test 3: statistical error . . . 20
  - 7.9.4 Test 4: computation time . . . 21

7.9.5 Test 5: cross-check on a full smile . . . 22
7.10 Variance reduction . . . 22
7.10.1 Principle and plan . . . 22
7.10.2 Method 1: conditional Monte Carlo . . . 23
7.10.3 Method 2: control variate . . . 23
7.10.4 Method 3: conditional Monte Carlo and control variate . . . 24
7.10.5 Results: comparison at the money . . . 26

8 Calibration **27**
8.1 The calibration problem . . . 27
8.2 Choice of the loss function . . . 27
8.3 Optimisation algorithm & Acceleration: a vectorised Lewis pricer . . . 28
8.4 Validation on a synthetic market . . . 29
8.4.1 Error measures in implied volatility. . . . 29
8.4.2 Test 1: recovery on exact prices . . . 30
8.4.3 Test 2: comparison of loss functions . . . 30
8.4.4 Test 3: quality of the fit . . . 30
8.5 Identifiability of the parameters . . . 31
8.6 Computation time . . . 31
8.7 Application to real data . . . 32

9 Conclusion **32**

1 Introduction and foundations of the Heston model

1.1 Motivation: going beyond the Black-Scholes framework

In the seminal model of Black, Scholes and Merton (1973), the price of the underlying asset $(S_t)_{t\geq 0}$ follows a geometric Brownian motion with constant volatility σ. This assumption leads to two consequences that are empirically contradicted by market data (especially since the 1987 crash):

- **The distribution of returns:** in reality, the empirical distribution of log-returns exhibits fat tails (*leptokurtosis*) and a marked asymmetry (*skewness*), whereas Black-Scholes assumes a normal distribution.
- **The volatility "smile":** if one inverts the Black-Scholes formula from quoted option prices for a given maturity, the implied volatility depends on the strike K. One observes a smile-shaped or downward-sloping curve (*skew*), which is incompatible with the assumption of constant volatility.

To capture these dynamics, Steven Heston proposed in 1993 a stochastic volatility model in which the instantaneous variance is no longer a static parameter, but a continuous mean-reverting random process.

1.2 Mathematical formulation of the model

Under the risk-neutral measure $\mathbb{Q}$, the dynamics are described by a system of two coupled stochastic differential equations:

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sqrt{v_t}\,S_t\,\mathrm{d}W_t^1, \tag{1}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^2, \tag{2}$$

with an instantaneous correlation between the two standard Brownian motions:

$$\mathrm{d}\langle W^1, W^2\rangle_t = \rho\,\mathrm{d}t, \qquad \rho \in [-1, 1]. \tag{3}$$

The instantaneous variance process $(v_t)_{t\geq 0}$ follows a Cox-Ingersoll-Ross (CIR) diffusion.

1.3 Financial interpretation of the parameters

The model depends on five structural parameters, each with a direct financial interpretation:

- $v_0 > 0$: **initial variance**. It is the level of spot instantaneous variance ($t = 0$).
- $\theta > 0$: **long-term mean level** (*long-term variance*). It is the level towards which the instantaneous variance is continuously pulled back.
- $\kappa > 0$: **speed of mean reversion** (*mean-reversion speed*). The higher κ, the faster v_t returns to θ after a shock. The half-life of a volatility shock is $t_{1/2} = \frac{\ln 2}{\kappa}$.
- $\xi > 0$: **volatility of the variance** (*vol of vol*). This parameter measures the uncertainty on the variance itself. On option prices, ξ controls the curvature of the smile (thickness of the tails of the return distribution).
- $\rho \in [-1, 1]$: **underlying/variance correlation**.
  - On equity markets, one empirically observes $\rho < 0$ (typically between -0.6 and -0.8). This is the **leverage effect** (Black, 1976): a fall in the underlying comes with a rise in volatility, which generates a pronounced negative *skew* (higher premiums on out-of-the-money puts).
  - When $\rho = 0$, the resulting smile is symmetric around the money.

1.4 Properties of the variance process and the Feller condition

The CIR variance process has two fundamental properties:

1. **No negative values:** the diffusion term $\sqrt{v_t}$ vanishes as $v_t \to 0$, which prevents the process from becoming negative.

2. **Feller condition:** for the boundary 0 to be inaccessible (that is, $v_t > 0$ almost surely for all $t > 0$, when $v_0 > 0$), the parameters must satisfy:

$$2\kappa\theta \geq \xi^2. \tag{4}$$

2 Decomposition of the call price and change of numéraire

We consider an arbitrage-free financial market with a risk-neutral probability $\mathbb{Q}$. The constant risk-free rate is denoted r, and the option maturity T.

The price at time $t = 0$ of a European call with strike K is the discounted expectation of its terminal payoff:

$$C = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_T - K)^+\big]. \tag{5}$$

Writing $(S_T - K)^+ = (S_T - K)\mathbf{1}_{\{S_T > K\}}$, we get:

$$\begin{aligned} C &= e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[S_T\mathbf{1}_{\{S_T>K\}}\big] - Ke^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[\mathbf{1}_{\{S_T>K\}}\big] \\ &= e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[S_T\mathbf{1}_{\{S_T>K\}}\big] - Ke^{-rT}P_2, \end{aligned} \tag{6}$$

where we set:

$$P_2 \coloneqq \mathbb{Q}(S_T > K) = \mathbb{E}^{\mathbb{Q}}\big[\mathbf{1}_{\{S_T>K\}}\big]. \tag{7}$$

Under $\mathbb{Q}$, the discounted process $(e^{-rt}S_t)_{t\geq 0}$ is a martingale. In particular:

$$\mathbb{E}^{\mathbb{Q}}[e^{-rT}S_T] = S_0 \iff \mathbb{E}^{\mathbb{Q}}\left[\frac{S_T}{S_0e^{rT}}\right] = 1. \tag{8}$$

The random variable $\frac{S_T}{S_0e^{rT}}$ is **positive with expectation** 1 **under** $\mathbb{Q}$. It therefore defines an admissible Radon-Nikodym density. We introduce the probability associated with the underlying as numéraire, denoted $\widetilde{\mathbb{Q}}$:

$$\frac{\mathrm{d}\widetilde{\mathbb{Q}}}{\mathrm{d}\mathbb{Q}} = \frac{S_T}{S_0e^{rT}}. \tag{9}$$

For any bounded measurable function g, the change-of-measure formula gives:

$$\mathbb{E}^{\widetilde{\mathbb{Q}}}[g(S_T)] = \mathbb{E}^{\mathbb{Q}}\left[\frac{\mathrm{d}\widetilde{\mathbb{Q}}}{\mathrm{d}\mathbb{Q}}\,g(S_T)\right] = \mathbb{E}^{\mathbb{Q}}\left[\frac{S_T}{S_0e^{rT}}\,g(S_T)\right]. \tag{10}$$

Applying it to the indicator $g(S_T) = \mathbf{1}_{\{S_T>K\}}$:

$$\mathbb{E}^{\widetilde{\mathbb{Q}}}\big[\mathbf{1}_{\{S_T>K\}}\big] = \frac{e^{-rT}}{S_0}\,\mathbb{E}^{\mathbb{Q}}\big[S_T\mathbf{1}_{\{S_T>K\}}\big] \iff e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[S_T\mathbf{1}_{\{S_T>K\}}\big] = S_0\,\widetilde{\mathbb{Q}}(S_T > K). \tag{11}$$

Setting $P_1 \coloneqq \widetilde{\mathbb{Q}}(S_T > K)$, the call price admits the structural decomposition:

$$C = S_0P_1 - Ke^{-rT}P_2, \qquad \text{with} \qquad \begin{cases} P_1 = \widetilde{\mathbb{Q}}(S_T > K), \\ P_2 = \mathbb{Q}(S_T > K). \end{cases} \tag{12}$$

3 Characteristic function of the log-price

3.1 Why use the characteristic function?

The decomposition (12) reduces the pricing of the call to the computation of two probabilities, P_1 and P_2. To obtain them, one would need to know the distribution of S_T under $\mathbb{Q}$ and under $\widetilde{\mathbb{Q}}$. The Heston model provides *no explicit density* for S_T: the variance (v_t) is itself random and correlated with the underlying, so that $\ln S_T$ follows a non-trivial mixture of distributions. One therefore cannot compute P_1 and P_2 directly by integrating a density.

We get around this difficulty by working in Fourier space. We introduce the log-price

$$X_t \coloneqq \ln S_t,$$

and its characteristic function under $\mathbb{Q}$:

$$\varphi(u) \coloneqq \mathbb{E}^{\mathbb{Q}}\left[e^{\mathrm{i}uX_T}\right], \qquad u \in \mathbb{R}. \tag{13}$$

This choice relies on three properties.

- **Existence.** We have $|e^{\mathrm{i}uX_T}| = 1$. The function φ is therefore always well defined, without any integrability assumption on X_T.
- **Characterisation.** The characteristic function entirely determines the distribution of X_T. All the information needed to compute P_1 and P_2 is therefore contained in it.
- **Computability.** In the Heston model, the *affine* structure of the dynamics makes it possible to obtain the characteristic function in quasi-closed form.

3.2 Dynamics of the log-price and intuition for the exponential-affine form

Itô's formula applied to $X_t = \ln S_t$ in (1) gives:

$$\mathrm{d}X_t = \left(r - \tfrac{1}{2}v_t\right)\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^1, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^2. \tag{14}$$

3.3 Feynman-Kac equation and exponential-affine form

For $t \in [0,T]$, we set:

$$f(t,x,v) \coloneqq \mathbb{E}^{\mathbb{Q}}\left[e^{\mathrm{i}uX_T} \,\middle|\, X_t = x,\ v_t = v\right].$$

By the Markov property, $f(t, X_t, v_t) = \mathbb{E}^{\mathbb{Q}}[e^{\mathrm{i}uX_T} \mid \mathcal{F}_t]$, which is a $\mathbb{Q}$-martingale: the $\mathrm{d}t$ term in Itô's formula is therefore zero. We thus obtain the Feynman-Kac PDE:

$$\partial_t f + \left(r - \tfrac{1}{2}v\right)\partial_x f + \kappa(\theta - v)\,\partial_v f + \tfrac{1}{2}v\,\partial_{xx} f + \rho\xi v\,\partial_{xv} f + \tfrac{1}{2}\xi^2 v\,\partial_{vv} f = 0, \tag{15}$$

with the terminal condition $f(T,x,v) = e^{\mathrm{i}ux}$.

Guided by the affine structure, we set $\tau \coloneqq T - t$ and look for a solution of exponential-affine form:

$$f(t,x,v) = \exp\big(C(\tau,u) + D(\tau,u)\,v + \mathrm{i}ux\big), \qquad C(0,u) = D(0,u) = 0. \tag{16}$$

The partial derivatives are:

$$\partial_t f = -(C' + D'v)\,f, \quad \partial_x f = \mathrm{i}u\,f, \quad \partial_{xx} f = -u^2 f, \quad \partial_v f = Df, \quad \partial_{vv} f = D^2 f, \quad \partial_{xv} f = \mathrm{i}uD\,f.$$

We substitute them into (15), then divide by $f \neq 0$:

$$-C' - D'v + \mathrm{i}u\left(r - \tfrac{1}{2}v\right) + \kappa\theta D - \kappa v D - \tfrac{1}{2}u^2 v + \mathrm{i}\rho\xi u\,vD + \tfrac{1}{2}\xi^2 v D^2 = 0.$$

This identity must hold for all $v > 0$.

$$\begin{cases} C'(\tau) = \mathrm{i}ur + \kappa\theta\, D(\tau), & (17) \\ D'(\tau) = \frac{1}{2}\xi^2 D(\tau)^2 - (\kappa - \mathrm{i}\rho\xi u)\, D(\tau) - \frac{1}{2}\left(u^2 + \mathrm{i}u\right). & (18) \end{cases}$$

Equation (18) is a **Riccati equation** with constant coefficients, which can be integrated explicitly. A simple integration of (17) then gives C.

3.4 Explicit solution

We introduce the complex quantities:

$$b(u) \coloneqq \kappa - \mathrm{i}\rho\xi u, \qquad d(u) \coloneqq \sqrt{b(u)^2 + \xi^2\,(u^2 + \mathrm{i}u)}, \qquad g(u) \coloneqq \frac{b(u) - d(u)}{b(u) + d(u)}. \tag{19}$$

The quadratic $\frac{1}{2}\xi^2 D^2 - bD - \frac{1}{2}(u^2 + \mathrm{i}u)$ has roots $(b \pm d)/\xi^2$. The classical method for solving Riccati equations then leads to:

$$D(\tau, u) = \frac{b - d}{\xi^2} \cdot \frac{1 - e^{-d\tau}}{1 - g\, e^{-d\tau}}, \tag{20}$$

$$C(\tau, u) = \mathrm{i}ur\tau + \frac{\kappa\theta}{\xi^2}\left[(b - d)\,\tau - 2\ln\left(\frac{1 - g\, e^{-d\tau}}{1 - g}\right)\right]. \tag{21}$$

One checks directly that $C(0) = D(0) = 0$, and that $C' = \mathrm{i}ur + \kappa\theta D$ thanks to the identity $g(b + d) = b - d$.

The characteristic function of X_T under $\mathbb{Q}$ is therefore:

$$\boxed{\varphi(u) = \exp\Big(C(T, u) + D(T, u)\, v_0 + \mathrm{i}u \ln S_0\Big).} \tag{22}$$

Conversely, the function f defined by (16), with C and D given by (20)–(21), is smooth and satisfies the PDE (15). By Itô's formula, $M_t = f(t, X_t, v_t)$ is a local martingale. For real u, the solution of the Riccati system is defined on the whole of $[0, T]$ and satisfies $\operatorname{Re} D(\tau, u) \leq 0$, so that M is bounded on $[0, T]$: it is a true martingale, and $M_t = \mathbb{E}^{\mathbb{Q}}[M_T \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[e^{\mathrm{i}uX_T} \mid \mathcal{F}_t]$. The exponential-affine form is therefore indeed the characteristic function, and not merely a candidate solution.

3.5 Characteristic functions under $\mathbb{Q}$ and under $\widetilde{\mathbb{Q}}$

The probability P_2 is defined under $\mathbb{Q}$. Its associated characteristic function is therefore directly $\varphi_2 \coloneqq \varphi$.

The probability P_1 is defined under $\widetilde{\mathbb{Q}}$. We then use the density (9) with $S_T = e^{X_T}$:

$$\varphi_1(u) \coloneqq \mathbb{E}^{\widetilde{\mathbb{Q}}}\left[e^{\mathrm{i}uX_T}\right] = \mathbb{E}^{\mathbb{Q}}\left[\frac{e^{X_T}}{S_0 e^{rT}}\, e^{\mathrm{i}uX_T}\right] = \frac{\mathbb{E}^{\mathbb{Q}}\left[e^{\mathrm{i}(u - \mathrm{i})X_T}\right]}{S_0 e^{rT}}.$$

By the martingale property (8), $\varphi(-\mathrm{i}) = \mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$. We deduce:

$$\varphi_1(u) = \frac{\varphi(u - \mathrm{i})}{\varphi(-\mathrm{i})}, \qquad \varphi_2(u) = \varphi(u). \tag{23}$$

4 Gil-Pelaez inversion formula

4.1 Statement and proof

The idea is to write the indicator $\mathbf{1}_{\{X>k\}}$ as an integral of trigonometric functions, then take the expectation. The basic tool is the Dirichlet integral: the function $x \mapsto \int_0^x \frac{\sin t}{t}\,\mathrm{d}t$ is bounded on $\mathbb{R}_+$ and converges to $\frac{\pi}{2}$ at $+\infty$. By the change of variable $t = au$, we deduce, for every real a:

$$\lim_{\substack{\varepsilon\to 0\\ A\to+\infty}} \int_\varepsilon^A \frac{\sin(au)}{u}\,\mathrm{d}u = \frac{\pi}{2}\operatorname{sgn}(a), \qquad \sup_{0<\varepsilon<A,\ a\in\mathbb{R}} \left|\int_\varepsilon^A \frac{\sin(au)}{u}\,\mathrm{d}u\right| \le M < +\infty. \tag{24}$$

Proposition 1 (Gil-Pelaez, 1951)**.** *Let X be an integrable real random variable with characteristic function $\phi(u) = \mathbb{E}[e^{\mathrm{i}uX}]$, and let $k \in \mathbb{R}$ be such that $\mathbb{P}(X = k) = 0$. Then:*

$$\mathbb{P}(X > k) = \frac{1}{2} + \frac{1}{\pi}\int_0^{+\infty} \operatorname{Re}\left[\frac{e^{-\mathrm{i}uk}\,\phi(u)}{\mathrm{i}u}\right]\mathrm{d}u. \tag{25}$$

Proof.

Step 1: Let $u > 0$. We have $\operatorname{Im}\left(e^{-\mathrm{i}uk}\phi(u)\right) = \mathbb{E}\big[\sin\left(u(X-k)\right)\big]$. The integrand of (25) is therefore:

$$h(u) := \frac{\mathbb{E}\big[\sin\left(u(X-k)\right)\big]}{u}.$$

Since $|\sin y| \le |y|$, we have $|h(u)| \le \mathbb{E}|X-k| < +\infty$. The function h is therefore bounded near 0, and the integral has no singularity at $u = 0$.

Step 2: Truncation and Fubini. Fix $0 < \varepsilon < A$. On $[\varepsilon, A] \times \Omega$, the function $(u, \omega) \mapsto \sin\left(u(X(\omega) - k)\right)/u$ is bounded by $1/\varepsilon$. It is therefore integrable for the product measure, and Fubini's theorem allows the interchange:

$$\int_\varepsilon^A h(u)\,\mathrm{d}u = \mathbb{E}\left[\int_\varepsilon^A \frac{\sin\left(u(X-k)\right)}{u}\,\mathrm{d}u\right] = \mathbb{E}\big[F_{\varepsilon,A}(X-k)\big],$$

where we denoted $F_{\varepsilon,A}(a) := \int_\varepsilon^A \frac{\sin(au)}{u}\,\mathrm{d}u$.

Step 3: passage to the limit. By (24), $F_{\varepsilon,A}(X-k) \to \frac{\pi}{2}\operatorname{sgn}(X-k)$ almost surely as $\varepsilon \to 0$ and $A \to +\infty$. Moreover, $|F_{\varepsilon,A}(X-k)| \le M$ uniformly. By dominated convergence:

$$\lim_{\substack{\varepsilon\to 0\\ A\to+\infty}} \int_\varepsilon^A h(u)\,\mathrm{d}u = \frac{\pi}{2}\,\mathbb{E}\big[\operatorname{sgn}(X-k)\big] = \frac{\pi}{2}\Big(\mathbb{P}(X>k) - \mathbb{P}(X<k)\Big).$$

Since $\mathbb{P}(X = k) = 0$, we have $\mathbb{P}(X < k) = 1 - \mathbb{P}(X > k)$. It follows that:

$$\int_0^{+\infty} \frac{\mathbb{E}\big[\sin\left(u(X-k)\right)\big]}{u}\,\mathrm{d}u = \frac{\pi}{2}\big(2\,\mathbb{P}(X>k) - 1\big).$$

This equality shows along the way that the integral converges, as an improper integral at $+\infty$. Isolating $\mathbb{P}(X > k)$, we obtain:

$$\mathbb{P}(X > k) = \frac{1}{2} + \frac{1}{\pi}\int_0^{+\infty} \operatorname{Re}\left[\frac{e^{-\mathrm{i}uk}\,\phi(u)}{\mathrm{i}u}\right]\mathrm{d}u$$

□

4.2 Application to the Heston model

Applying (25), we obtain for $j \in \{1, 2\}$:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^{+\infty} \mathrm{Re}\left[\frac{e^{-\mathrm{i}u \ln K}\, \varphi_j(u)}{\mathrm{i}u} \right] \mathrm{d}u, \qquad \varphi_1(u) = \frac{\varphi(u - \mathrm{i})}{\varphi(-\mathrm{i})}, \quad \varphi_2(u) = \varphi(u). \tag{26}$$

Substituting into the decomposition (12), the price of the European call is:

$$\boxed{C = S_0 P_1 - K e^{-rT} P_2.} \tag{27}$$

This is called a *semi-closed-form* formula: all the ingredients are explicit, except for an integral over $\mathbb{R}_+$.

5 Lewis formula

5.1 Motivation: the limitations of the Gil-Pelaez formula

Formula (26) is exact, but it has three numerical drawbacks.

- **Two integrals.** P_1 and P_2 must be computed separately, then combined.
- **Slow decay.** The integrand behaves like $|\varphi_j(u)|/u$: the factor $1/u$ slows down convergence at $+\infty$ and requires a high truncation bound.
- **Behaviour at $u = 0$.** The integrand remains bounded, but it is written as a quotient of two quantities that tend to 0. Numerically, this causes losses of precision near the origin.

The Lewis formula (2001) gets around these three difficulties. It expresses the call price directly through *a single* integral, whose integrand decays like $1/u^2$ and remains regular at $u = 0$.

With Gil-Pelaez, we first decomposed the price into two probabilities, then inverted each of them. Lewis's idea is more direct: one writes the *payoff itself* as a Fourier integral, then takes the expectation. The characteristic function then appears immediately.

One difficulty arises: the payoff $x \mapsto (e^x - K)^+$ grows exponentially, so it has no usual Fourier transform. We get around this in two steps.

- We write $(e^x - K)^+ = e^x - \min(e^x, K)$. The first term is priced directly: $e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T] = S_0$. The second is a function bounded by K.
- We evaluate the Fourier transform at a *complex* argument $z = u + \mathrm{i}\nu$. This multiplies the function by a damping factor $e^{-\nu x}$, which makes it integrable.

5.2 Statement and proof

We adopt the convention $\hat{f}(u) = \int_{\mathbb{R}} e^{\mathrm{i}ux} f(x)\, \mathrm{d}x$, consistent with that of the characteristic function. We recall the inversion theorem: if f is continuous and integrable, and if $\hat{f}$ is integrable, then for all x, $f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-\mathrm{i}ux} \hat{f}(u)\, \mathrm{d}u$.

Proposition 2 (Lewis, 2001)**.** *In the Heston model, the price of the European call is:*

$$\boxed{C = S_0 - \frac{\sqrt{K}\, e^{-rT}}{\pi} \int_0^{+\infty} \mathrm{Re}\Big[e^{-\mathrm{i}u \ln K}\, \varphi\big(u - \tfrac{\mathrm{i}}{2}\big) \Big] \frac{\mathrm{d}u}{u^2 + \frac{1}{4}}.} \tag{28}$$

Proof.

Step 1: Decomposition of the payoff. We set $h(x) \coloneqq \min(e^x, K)$. Since $(e^x - K)^+ = e^x - h(x)$ and $e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T] = S_0$, we have:

$$C = S_0 - e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[h(X_T)\big].$$

Step 2: Generalised Fourier transform. For $z \in \mathbb{C}$ such that $0 < \mathrm{Im}(z) < 1$, we compute:

$$\hat{h}(z) \coloneqq \int_{\mathbb{R}} e^{\mathrm{i}zx} h(x)\,\mathrm{d}x = \int_{-\infty}^{\ln K} e^{(\mathrm{i}z+1)x}\,\mathrm{d}x + K\int_{\ln K}^{+\infty} e^{\mathrm{i}zx}\,\mathrm{d}x.$$

The first integral converges since $\mathrm{Re}(\mathrm{i}z+1) = 1-\mathrm{Im}(z) > 0$, the second since $\mathrm{Re}(\mathrm{i}z) = -\,\mathrm{Im}(z) < 0$. We obtain:

$$\hat{h}(z) = \frac{K^{\mathrm{i}z+1}}{\mathrm{i}z+1} - \frac{K^{\mathrm{i}z+1}}{\mathrm{i}z} = \frac{K^{1+\mathrm{i}z}}{z^2 - \mathrm{i}z}.$$

We place ourselves in the middle of the strip, on the line $\mathrm{Im}(z) = \frac{1}{2}$, by setting $z = u + \frac{\mathrm{i}}{2}$. A direct computation gives $z^2 - \mathrm{i}z = u^2 + \frac{1}{4}$ and $K^{1+\mathrm{i}z} = \sqrt{K}\,e^{\mathrm{i}u\ln K}$. Hence:

$$\hat{h}(u + \tfrac{\mathrm{i}}{2}) = \frac{\sqrt{K}\,e^{\mathrm{i}u\ln K}}{u^2 + \frac{1}{4}}.$$

Step 3: inversion. We set $f(x) \coloneqq e^{-x/2}h(x) = \min\left(e^{x/2}, Ke^{-x/2}\right)$. This function is continuous and integrable on $\mathbb{R}$, since it decays exponentially on both sides. Its Fourier transform is $\hat{f}(u) = \hat{h}(u + \frac{\mathrm{i}}{2})$, which is also continuous and integrable on $\mathbb{R}$ since $\hat{f}(u) = O(\frac{1}{u^2})$. The inversion theorem applies. Multiplying by $e^{x/2}$, we obtain for all x:

$$h(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\mathrm{i}(u+\mathrm{i}/2)x}\,\frac{\sqrt{K}\,e^{\mathrm{i}u\ln K}}{u^2 + \frac{1}{4}}\,\mathrm{d}u.$$

Step 4: Expectation and Fubini. We replace x by X_T and take the expectation. The integrand is bounded in modulus by $\frac{e^{\frac{X_T}{2}}\sqrt{K}}{u^2+\frac{1}{4}}$. Now, by Jensen's inequality, $\mathbb{E}^{\mathbb{Q}}[e^{X_T/2}] = \mathbb{E}^{\mathbb{Q}}[\sqrt{S_T}] \le \sqrt{\mathbb{E}^{\mathbb{Q}}[S_T]} < +\infty$, and $u \mapsto 1/(u^2 + \frac{1}{4})$ is integrable on $\mathbb{R}$. Fubini's theorem therefore allows the interchange:

$$\mathbb{E}^{\mathbb{Q}}\big[h(X_T)\big] = \frac{\sqrt{K}}{2\pi}\int_{\mathbb{R}} \frac{e^{\mathrm{i}u\ln K}\,\varphi(-u-\frac{\mathrm{i}}{2})}{u^2+\frac{1}{4}}\,\mathrm{d}u = \frac{\sqrt{K}}{2\pi}\int_{\mathbb{R}} G(u)\,\mathrm{d}u,$$

with $G(u) \coloneqq e^{-\mathrm{i}u\ln K}\,\varphi(u - \frac{\mathrm{i}}{2})/(u^2 + \frac{1}{4})$. The second equality is obtained by the change of variable $u \mapsto -u$.

Step 5: reduction to $\mathbb{R}_+$. Since X_T is real, we have $\overline{\varphi(u - \frac{\mathrm{i}}{2})} = \mathbb{E}^{\mathbb{Q}}[e^{-\mathrm{i}uX_T}e^{X_T/2}] = \varphi(-u - \frac{\mathrm{i}}{2})$. We deduce $G(-u) = \overline{G(u)}$, hence $\int_{\mathbb{R}} G(u)\,\mathrm{d}u = 2\int_0^{+\infty} \mathrm{Re}\,[G(u)]\,\mathrm{d}u$. Substituting into Step 1, we obtain (28). □

5.3 Comparison with Gil-Pelaez

- **A single integral**
- **No difficulty at** $u = 0$
- **Decay in** $1/u^2$, instead of $1/u$: the integral can be truncated at a much lower bound for the same accuracy.

6 Numerical validation

The Gil-Pelaez and Lewis formulas were implemented in Python.

6.1 Numerical setting

Reference parameters. Unless stated otherwise, we use the parameters of Table 1. We choose $v_0 = \theta$, for a reason explained in Section 6.4. The Feller condition (4) is not satisfied: $2\kappa\theta = 0.16 < \xi^2 = 0.25$. This does not affect the pricing formulas, which remain valid in this regime.

Parameter	S_0	K	T	r	v_0	κ	θ	ξ	ρ
Value	100	100	1	0.05	0.04	2	0.04	0.5	-0.7

Table 1: Reference parameters.

6.2 Checks on the characteristic function

Two exact identities, which follow directly from the definitions, provide a first check of the implementation of φ:

- $\varphi(0) = \mathbb{E}^{\mathbb{Q}}[1] = 1$, since φ is a characteristic function;
- $\varphi(-\mathrm{i}) = \mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$, by the martingale property (8).

Numerically, we obtain $\varphi(0) = 1.0000000000$ and $\varphi(-\mathrm{i}) = 105.1271096376$, which matches $S_0 e^{rT}$ on all ten displayed decimals.

6.3 Consistency between the Gil-Pelaez and Lewis formulas

The two formulas are exact expressions of the same price, obtained by different routes. Table 2 compares the prices obtained for several strikes.

K	Gil-Pelaez	Lewis	Absolute difference
80	25.23676408	25.23676411	3.1×10^{-8}
90	17.14005510	17.14005511	1.8×10^{-8}
100	10.15462701	10.15462703	1.3×10^{-8}
110	4.86618257	4.86618258	1.3×10^{-8}
120	1.73823234	1.73823236	1.9×10^{-8}

Table 2: Call prices obtained with the Gil-Pelaez and Lewis formulas.

The differences are all of order 10^{-8}. They are entirely explained by the truncation of the integral and, for Gil-Pelaez, by its starting point at ε.

6.4 Black-Scholes limit as $\xi \to 0$

The expected limiting price. When $\xi = 0$, the random term disappears from the variance equation (2), which becomes an ordinary differential equation:

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t \qquad \Longrightarrow \qquad v_t = \theta + (v_0 - \theta)\,e^{-\kappa t}.$$

With $v_0 = \theta$, the variance remains constant, equal to θ. The underlying then follows a geometric Brownian motion with volatility $\sigma = \sqrt{\theta}$, that is, the Black-Scholes model. The Heston price must therefore converge to the Black-Scholes price with volatility $\sqrt{\theta}$.

Remark 1. If $v_0 \neq \theta$, the limiting variance is deterministic but time-dependent. The limiting price is then the Black-Scholes price with average volatility $\sigma_{\text{eff}}^2 = \frac{1}{T}\int_0^T v_t \, \mathrm{d}t = \theta + (v_0 - \theta)\frac{1-e^{-\kappa T}}{\kappa T}$, and not $\sqrt{\theta}$. The choice $v_0 = \theta$ allows a direct comparison with $\sigma = \sqrt{\theta}$.

Direct test. With $\xi = 10^{-4}$, $\rho = 0$ and the other parameters of Table 1, we obtain:

Method	Call price
Black-Scholes, $\sigma = \sqrt{\theta}$	10.450584
Heston, Gil-Pelaez	10.450584
Heston, Lewis	10.450583

Flattening of the smile. A more visual test consists in following the implied volatility smile as ξ decreases. For each strike K, we compute the Heston price with the Lewis formula, then deduce the implied volatility, that is, the volatility σ that reproduces this price in the Black-Scholes formula. Since the Black-Scholes price is strictly increasing in σ, this volatility is unique; we compute it with Brent's method. If the Heston model reduces to Black-Scholes, we must obtain the same volatility for all strikes.

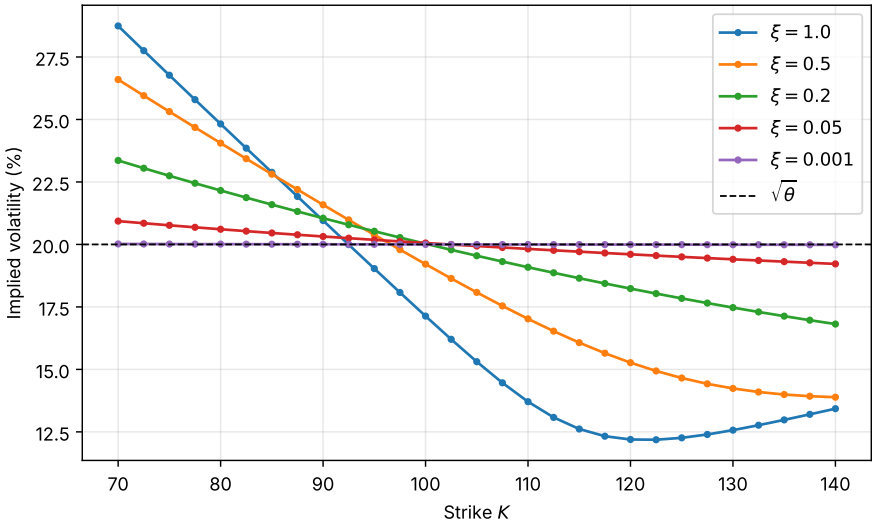


Figure 1: Implied volatility smile of the Heston model for different values of ξ ($\rho = -0.7$, $T = 1$, other parameters from Table 1). As $\xi \to 0$, the smile flattens towards the line $\sigma = \sqrt{\theta}$.

Figure 1 confirms the two expected effects.

- **For large** ξ, we observe a pronounced negative *skew*: implied volatilities are higher for low strikes. This is the leverage effect induced by $\rho < 0$, described in Section 1.
- **As** ξ **decreases**, the smile gradually flattens. For $\xi = 0.001$, it coincides with the line $\sigma = \sqrt{\theta} = 20\,\%$: without randomness in the variance, the correlation ρ no longer has any effect, and the model reduces to Black-Scholes.

6.5 Summary

The three tests are conclusive.

- The characteristic function satisfies the exact identities $\varphi(0) = 1$ and $\varphi(-\mathrm{i}) = S_0 e^{rT}$.
- The Gil-Pelaez and Lewis formulas give the same price, up to the accuracy of the quadrature.

- The Heston model converges to the Black-Scholes model as $\xi \to 0$, both in price and in implied volatility.

These semi-closed-form formulas can therefore serve as reference prices to validate the Monte Carlo methods.

7 Monte Carlo simulation

7.1 Why simulate when a semi-closed-form formula is available?

For a European option, the Lewis formula (28) gives a price accurate to 10^{-8} in a few milliseconds. No Monte Carlo method can compete on this ground. Simulation meets two other needs.

- **An independent check.** The Lewis price relies on a chain of analytical results, notably the characteristic function. Monte Carlo uses none of them: it directly simulates the dynamics (1)–(2). Agreement between the two methods therefore validates both the theory and the two implementations.
- **Products without a closed-form formula.** Barrier, Asian or autocallable options depend on the whole path of the underlying. Even in the Heston model, their price has no simple Fourier representation. Simulation is then the standard tool.

In this section, we focus on the European call, whose exact price is known. This is precisely what makes it possible to measure the error of the simulation schemes, before using them where no reference exists.

7.2 Principle of the Monte Carlo estimator

Ideal estimator. Let $S_T^{(1)}, \dots, S_T^{(N)}$ be independent copies of S_T under $\mathbb{Q}$. We estimate the price $C = e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$ by the empirical mean

$$\widehat{C}_N = \frac{1}{N}\sum_{i=1}^{N} Y_i, \qquad Y_i := e^{-rT}\big(S_T^{(i)} - K\big)^+.$$

Since $0 \le Y_i \le e^{-rT} S_T^{(i)}$ and $\mathbb{E}^{\mathbb{Q}}[S_T] < +\infty$, the Y_i are integrable: the strong law of large numbers gives $\widehat{C}_N \to C$ almost surely. If moreover $\sigma^2 := \mathrm{Var}(Y_1) < +\infty$, the central limit theorem gives

$$\sqrt{N}\,(\widehat{C}_N - C) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \sigma^2).$$

Replacing σ by the empirical standard deviation $\widehat{\sigma}_N$ (Slutsky's lemma), we deduce the asymptotic 95 % confidence interval:

$$\left[\ \widehat{C}_N - 1.96\,\frac{\widehat{\sigma}_N}{\sqrt{N}}\ ;\ \widehat{C}_N + 1.96\,\frac{\widehat{\sigma}_N}{\sqrt{N}}\ \right]. \tag{29}$$

The statistical error therefore decreases as $N^{-1/2}$, whatever the dimension of the problem: dividing the error by 10 costs 100 times more paths.

Actual estimator: bias and variance. We do not know how to simulate S_T exactly in an efficient way. Instead, we simulate an approximation $\widehat{S}_T^{(n)}$ obtained by a scheme with n time steps $\Delta = T/n$. The estimator $\widehat{C}_{N,n}$ then makes two errors of different natures, which are separated by the decomposition of the mean squared error **MSE**:

$$\mathbb{E}\Big[\big(\widehat{C}_{N,n} - C\big)^2\Big] = \underbrace{\Big(\mathbb{E}\big[\widehat{Y}^{(n)}\big] - C\Big)^2}_{\text{bias}^2\ \text{(discretisation)}} + \underbrace{\frac{\mathrm{Var}\,\big(\widehat{Y}^{(n)}\big)}{N}}_{\text{statistical error}}. \tag{30}$$

The bias, or *weak error* of the scheme, depends only on n; it does not vanish by increasing N. The statistical error depends only on N. A good scheme is therefore one whose bias becomes negligible compared with the confidence interval with *few* time steps. This is the criterion that will guide the whole comparison.

7.3 The Euler scheme and the positivity problem

Naive scheme. We discretise the log-price $X_t = \ln S_t$ rather than S_t, which guarantees $S_t > 0$. The Euler scheme applied to (14) and (2) reads:

$$v_{t+\Delta} = v_t + \kappa(\theta - v_t)\Delta + \xi\sqrt{v_t\Delta}\, Z_v,$$
$$X_{t+\Delta} = X_t + \left(r - \tfrac{1}{2}v_t\right)\Delta + \sqrt{v_t\Delta}\, Z_S,$$

where Z_v and $Z_\perp$ are two independent standard Gaussians, and $Z_S = \rho Z_v + \sqrt{1-\rho^2}\, Z_\perp$.

This scheme does not preserve positivity. Given v_t, the variable $v_{t+\Delta}$ is Gaussian, hence negative with strictly positive probability. At the next step, the square root $\sqrt{v_{t+\Delta}}$ is no longer defined. The problem is structural, and it is all the more frequent as v_t is close to 0, that is, precisely when the Feller condition is violated.

Possible corrections. Several ways of handling negative values have been proposed: absorption ($v \leftarrow v^+$), reflection ($v \leftarrow |v|$), partial or full truncation. Lord, Koekkoek and Van Dijk (2010) compare them systematically; **full truncation** gives the smallest bias. It consists in replacing v_t by $v_t^+ = \max(v_t, 0)$ in all the coefficients, *without* modifying the state itself:

$$v_{t+\Delta} = v_t + \kappa(\theta - v_t^+)\Delta + \xi\sqrt{v_t^+\Delta}\, Z_v, \tag{31}$$
$$X_{t+\Delta} = X_t + \left(r - \tfrac{1}{2}v_t^+\right)\Delta + \sqrt{v_t^+\Delta}\, Z_S. \tag{32}$$

When $v_t < 0$, the variance is "frozen" at 0 in the diffusion, and only the drift $\kappa\theta\Delta > 0$ brings it back towards positive values.

Limitations. For smooth coefficients, the Euler scheme has weak order 1: its bias decreases as $O(\Delta)$. Here, the function $v \mapsto \sqrt{v}$ is not Lipschitz at 0, and the truncation introduces an additional error each time $v_t < 0$. When the Feller condition is violated, the proportion of affected steps becomes considerable (up to 58 % in our tests), and the bias becomes very large for a moderate number of steps. To do better, one must exploit the exact law of the CIR process.

7.4 The exact law of the CIR process

7.4.1 Conditional moments

Proposition 3. *For all $\Delta > 0$, conditionally on v_t:*

$$m \coloneqq \mathbb{E}[v_{t+\Delta} \mid v_t] = \theta + (v_t - \theta)\, e^{-\kappa\Delta}, \tag{33}$$
$$s^2 \coloneqq \mathrm{Var}(v_{t+\Delta} \mid v_t) = \frac{v_t\, \xi^2 e^{-\kappa\Delta}}{\kappa}(1 - e^{-\kappa\Delta}) + \frac{\theta\xi^2}{2\kappa}(1 - e^{-\kappa\Delta})^2. \tag{34}$$

Proof. By the Markov property, we may assume $t = 0$ and v_0 deterministic. We denote $m(s) = \mathbb{E}[v_s]$ and $q(s) = \mathbb{E}[v_s^2]$.

First moment. In integral form, $v_s = v_0 + \int_0^s \kappa(\theta - v_u)\,\mathrm{d}u + \xi\int_0^s \sqrt{v_u}\,\mathrm{d}W_u^2$. Since $\mathbb{E}\int_0^s v_u\,\mathrm{d}u < +\infty$, the stochastic integral is a true martingale, with zero mean. We obtain $m'(s) = \kappa(\theta - m(s))$, with $m(0) = v_0$, hence $m(s) = \theta + (v_0 - \theta)e^{-\kappa s}$.

Second moment. Itô's formula applied to v_s^2 gives $\mathrm{d}(v_s^2) = 2v_s\,\mathrm{d}v_s + \xi^2 v_s\,\mathrm{d}s = [(2\kappa\theta+\xi^2)v_s - 2\kappa v_s^2]\mathrm{d}s + 2\xi v_s^{3/2}\,\mathrm{d}W_s^2$. Taking expectations (the martingale term vanishes, since all moments of the CIR process are finite), we obtain the linear equation $q'(s) = (2\kappa\theta+\xi^2)\,m(s) - 2\kappa\,q(s)$. We solve it by variation of constants, then compute $q(\Delta) - m(\Delta)^2$, which gives (34). □

7.4.2 Laplace transform and noncentral chi-squared distribution

One can go much further than the first two moments: the conditional law of $v_{t+\Delta}$ is known explicitly. The proof uses *exactly the same method* as for the characteristic function in Section 3: a solution of exponential-affine form of a Feynman-Kac equation, which leads to a Riccati equation.

Proposition 4 (CIR transition law)**.** *For all $\Delta > 0$ and all $s \geq 0$:*

$$\mathbb{E}\big[e^{-sv_{t+\Delta}} \mid v_t\big] = \Big(1 + \tfrac{\xi^2 s}{2\kappa}\big(1-e^{-\kappa\Delta}\big)\Big)^{-\frac{2\kappa\theta}{\xi^2}} \exp\left(\frac{-s\,v_t\,e^{-\kappa\Delta}}{1+\frac{\xi^2 s}{2\kappa}\big(1-e^{-\kappa\Delta}\big)}\right). \tag{35}$$

Consequently, with

$$c = \frac{\xi^2\big(1-e^{-\kappa\Delta}\big)}{4\kappa}, \qquad d = \frac{4\kappa\theta}{\xi^2}, \qquad \lambda = \frac{v_t\,e^{-\kappa\Delta}}{c}, \tag{36}$$

we have $v_{t+\Delta} \mid v_t \overset{d}{=} c\,\chi'^2_d(\lambda)$, where $\chi'^2_d(\lambda)$ denotes the noncentral chi-squared distribution with d degrees of freedom and noncentrality parameter λ.

Proof. Step 1: exponential-affine form. We fix $s \geq 0$ and set $u(t,v) = \mathbb{E}[e^{-sv_T} \mid v_t = v]$. As for the characteristic function, $u(t,v_t)$ is a martingale, and u satisfies the Feynman-Kac PDE

$$\partial_t u + \kappa(\theta - v)\,\partial_v u + \tfrac{1}{2}\xi^2 v\,\partial_{vv} u = 0, \qquad u(T,v) = e^{-sv}.$$

Since the drift and the instantaneous variance are affine in v, we look for $u = \exp\big(A(\tau) + B(\tau)\,v\big)$ with $\tau = T-t$, $A(0) = 0$ and $B(0) = -s$. Substituting and identifying the constant term and the coefficient of v, we obtain:

$$A'(\tau) = \kappa\theta\,B(\tau), \qquad B'(\tau) = -\kappa\,B(\tau) + \tfrac{1}{2}\xi^2 B(\tau)^2.$$

Step 2: solving the Riccati equation. The equation for B is a Bernoulli equation. Since $B(0) = -s < 0$, B does not vanish, and $w := 1/B$ satisfies the linear equation $w' = -B'/B^2 = \kappa w - \frac{1}{2}\xi^2$, with $w(0) = -1/s$. We deduce $w(\tau) = -e^{\kappa\tau}\big[\frac{1}{s} + \frac{\xi^2}{2\kappa}\big(1-e^{-\kappa\tau}\big)\big]$, then

$$B(\tau) = \frac{-s\,e^{-\kappa\tau}}{1+\frac{\xi^2 s}{2\kappa}\big(1-e^{-\kappa\tau}\big)}, \qquad A(\tau) = \kappa\theta\int_0^\tau B = -\frac{2\kappa\theta}{\xi^2}\ln\Big(1+\tfrac{\xi^2 s}{2\kappa}\big(1-e^{-\kappa\tau}\big)\Big).$$

Differentiating the expression of A indeed gives back $\kappa\theta B$. With $\tau = \Delta$, we obtain (35).

Step 3: identification of the law. The Laplace transform of $Y = c\,\chi'^2_d(\lambda)$ is $\mathbb{E}[e^{-sY}] = (1+2cs)^{-d/2}\exp\big(-\lambda cs/(1+2cs)\big)$. With the parameters (36), we have $2c = \frac{\xi^2}{2\kappa}\big(1-e^{-\kappa\Delta}\big)$, $\frac{d}{2} = \frac{2\kappa\theta}{\xi^2}$ and $\lambda c = v_t e^{-\kappa\Delta}$: the two transforms coincide for all $s \geq 0$. Now, the Laplace transform characterises the law of a positive random variable. Hence the equality in law. □

7.4.3 The parameter d and the Feller condition

The number of degrees of freedom $d = 4\kappa\theta/\xi^2$ depends only on the model parameters, and

$$d \geq 2 \iff 2\kappa\theta \geq \xi^2.$$

The Feller condition (4) is therefore nothing other than the condition $d \geq 2$, expressed in the language of the noncentral chi-squared distribution. It has a direct translation on the shape of the transition law. Near 0, the density of $\chi'^2_d(\lambda)$ behaves like $x^{d/2-1}$, up to a constant:

- if $d > 2$ (Feller satisfied), the density tends to 0 at 0: the variance stays away from 0;
- if $d < 2$ (Feller violated), the density *blows up* at 0: a large part of the mass is concentrated in a tiny neighbourhood of 0.

7.5 Why avoid the exact law in the simulation?

Proposition 4 makes it possible to draw $v_{t+\Delta}$ exactly, and this draw is cheap: we actually use it as the reference in Test 1. The difficulty does not lie in the variance, but in the log-price. By (38), advancing X_t exactly also requires drawing the integral $\int_t^{t+\Delta} v_s \, \mathrm{d}s$ *conditionally* on v_t and $v_{t+\Delta}$. The law of this integral is only known through its characteristic function, which involves modified Bessel functions. The exact method of Broadie and Kaya (2006) inverts it numerically for each path and at each step: it is unbiased, but far too slow for hundreds of thousands of paths.

In practice, this integral is therefore approximated, for instance by the trapezoidal rule. From then on, drawing $v_{t+\Delta}$ exactly no longer eliminates the discretisation error, and the gain in accuracy becomes marginal. The QE scheme is then preferred for a practical reason: each step uses only one Gaussian or uniform variable, transformed by an explicit formula, which vectorises directly over all paths. The exact draw of the noncentral chi-squared, on the contrary, requires a random number of draws (a Poisson variable, then a Gamma variable), which is more costly.

7.6 Andersen's QE scheme for the variance

At each step, we know m and s^2 exactly from Proposition 3. Andersen's idea is to replace the exact law $c\,\chi'^2_d(\lambda)$ by a law that is *simpler to draw*, chosen in a parametric family whose shape mimics the exact law, and whose parameters are fitted to reproduce m and s^2 exactly. The choice of family depends on the regime, measured by the dimensionless ratio

$$\psi := \frac{s^2}{m^2}.$$

What ψ measures, and what it depends on. The quantity $\sqrt{\psi} = s/m$ is the coefficient of variation of the step: the standard deviation of $v_{t+\Delta}$ relative to its mean. Since $v_{t+\Delta} \geq 0$, a high value of ψ can only be obtained with a lot of mass near 0 and a long right tail. As a benchmark, the exponential distribution satisfies exactly $\psi = 1$. Two limiting cases shed light on the role of the parameters.

- **Small step, non-zero variance.** For $\kappa\Delta \ll 1$, we have $e^{-\kappa\Delta} \approx 1$ and $1 - e^{-\kappa\Delta} \approx \kappa\Delta$. Formulas (33)–(34) then give $m \approx v_t$ and $s^2 \approx \xi^2 v_t \Delta$. We recover:

  $$\psi \approx \frac{\xi^2 \Delta}{v_t}.$$

  The parameter ψ therefore increases with the step Δ and with the vol of vol ξ, and decreases with the variance level v_t.

- **Zero variance.** At $v_t = 0$, the exact computation gives $m = \theta(1 - e^{-\kappa\Delta})$ and $s^2 = \frac{\theta\xi^2}{2\kappa}(1 - e^{-\kappa\Delta})^2$, hence

  $$\psi = \frac{\xi^2}{2\kappa\theta} = \frac{2}{d},$$

  independently of Δ, where d is the number of degrees of freedom (36).

This second case makes the link with the Feller condition. If it is satisfied ($d > 2$), we have $\psi < 1$ near 0: the quadratic branch suffices. If it is violated, ψ becomes large as soon as the

variance approaches 0 ($\psi = 18$ for $d = 0.11$): the scheme then switches to the exponential branch, precisely where the exact law concentrates its mass near 0.

A small ψ corresponds to a law concentrated around its mean, far from 0 (large noncentrality parameter λ). A large ψ corresponds to a law with a lot of mass near 0 (small λ, small d).

7.6.1 Quadratic branch (small ψ)

Justification of the form. A noncentral chi-squared with one degree of freedom can be written exactly as $(\sqrt{\lambda} + Z)^2$, with $Z \sim \mathcal{N}(0, 1)$. When λ is large, the law $\chi'^2_d(\lambda)$ is dominated by its noncentral component, and it is well approximated by a law of this form, suitably rescaled. We therefore set

$$v_{t+\Delta} = a\,(b + Z_v)^2, \qquad Z_v \sim \mathcal{N}(0, 1), \quad a > 0,\ b \geq 0.$$

Moment matching. We have $\mathbb{E}[(b + Z)^2] = 1 + b^2$ and $\mathbb{E}[(b + Z)^4] = b^4 + 6b^2 + 3$, hence

$$\mathbb{E}[v_{t+\Delta}] = a(1 + b^2), \qquad \mathrm{Var}(v_{t+\Delta}) = a^2\big(b^4 + 6b^2 + 3 - (1 + b^2)^2\big) = a^2(2 + 4b^2).$$

Imposing $\mathbb{E}[v_{t+\Delta}] = m$ and $\mathrm{Var}(v_{t+\Delta}) = s^2$, then dividing, we obtain an equation in $x = b^2$ alone:

$$\psi = \frac{2 + 4x}{(1 + x)^2} \iff \psi x^2 + (2\psi - 4)x + (\psi - 2) = 0.$$

The reduced discriminant is $(\psi - 2)^2 - \psi(\psi - 2) = 2(2 - \psi)$. It is non-negative if and only if $\psi \leq 2$, and the largest root is:

$$b^2 = \frac{2}{\psi} - 1 + \sqrt{\frac{2}{\psi}}\sqrt{\frac{2}{\psi} - 1}\ \geq 0, \qquad a = \frac{m}{1 + b^2}. \tag{37}$$

The quadratic branch is therefore only defined for $\psi \leq 2$.

7.6.2 Exponential branch (large ψ)

Justification of the form. The noncentral chi-squared is a Poisson mixture of central chi-squared distributions: $\chi'^2_d(\lambda) \overset{d}{=} \chi^2_{d+2P}$, with $P \sim \mathcal{P}(\lambda/2)$. When λ is small, $P = 0$ with high probability, and the law is close to a central χ^2_d. For small d, this law concentrates a large mass near 0; for $d = 2$, it is exactly an exponential distribution. We therefore approximate the exact law by a mixture of a mass at 0 and an exponential distribution:

$$\mathbb{Q}(v_{t+\Delta} = 0) = p, \qquad \mathbb{Q}(v_{t+\Delta} > x) = (1 - p)\,e^{-\beta x} \quad (x \geq 0).$$

Moment matching. For this mixture, $\mathbb{E}[v_{t+\Delta}] = (1 - p)/\beta$ and $\mathbb{E}[v^2_{t+\Delta}] = 2(1 - p)/\beta^2$. Equating the means gives $\beta = (1 - p)/m$. It then follows that

$$\psi = \frac{\mathbb{E}[v^2_{t+\Delta}]}{m^2} - 1 = \frac{2}{1 - p} - 1, \qquad \text{hence} \qquad p = \frac{\psi - 1}{\psi + 1}, \qquad \beta = \frac{1 - p}{m}.$$

We have $p \in [0, 1)$ if and only if $\psi \geq 1$: the exponential branch is only defined for $\psi \geq 1$.

Sampling by inversion. The cumulative distribution function of the mixture is $F(x) = p + (1 - p)(1 - e^{-\beta x})$ for $x \geq 0$. With $U \sim \mathcal{U}(0, 1)$, the inversion method gives:

$$v_{t+\Delta} = F^{-1}(U) = \begin{cases} 0 & \text{if } U \leq p, \\ \dfrac{1}{\beta}\ln\!\left(\dfrac{1 - p}{1 - U}\right) & \text{if } U > p. \end{cases}$$

7.6.3 Choice of branch and properties

The two branches coexist on the interval $\psi \in [1, 2]$. Andersen proposes to switch at a threshold ψ_c chosen in this interval; the value $\psi_c = 1.5$ is the usual choice, and the results are not very sensitive to it. We therefore retain:

$$\psi \leq \psi_c \;\Rightarrow\; \text{quadratic branch}, \qquad \psi > \psi_c \;\Rightarrow\; \text{exponential branch}.$$

The QE scheme then has three properties that the Euler scheme lacks.

- **Positivity.** By construction, $v_{t+\Delta} \geq 0$ in both branches: no truncation is needed.
- **Exact moments.** The conditional mean and variance are those of the exact law, whatever the step Δ. In particular, $\mathbb{E}[v_T]$ has no discretisation bias.
- **Justified form.** In each regime, the chosen family is the limiting form of the true law, and not an arbitrary law.

QE is nevertheless not an exact simulation: only the limiting form is reproduced. For example, the point mass at 0 of the exponential branch does not exist in the exact law, which is continuous for $d > 0$; it approximates the very strong concentration near 0 observed when $d < 2$.

7.7 Discretisation of the log-price

Elimination of the variance Brownian motion. We decompose $W^1 = \rho W^2 + \sqrt{1-\rho^2}\, W^\perp$, with $W^\perp$ a Brownian motion independent of W^2. Integrating (14) over $[t, t+\Delta]$:

$$X_{t+\Delta} = X_t + r\Delta - \frac{1}{2}\int_t^{t+\Delta} v_s \,\mathrm{d}s + \rho \int_t^{t+\Delta} \sqrt{v_s}\,\mathrm{d}W_s^2 + \sqrt{1-\rho^2}\int_t^{t+\Delta} \sqrt{v_s}\,\mathrm{d}W_s^\perp.$$

The integral with respect to W^2 is difficult to simulate directly. But the variance equation, integrated over the same interval, makes it possible to express it in terms of quantities that have already been simulated:

$$v_{t+\Delta} = v_t + \kappa\theta\Delta - \kappa\int_t^{t+\Delta} v_s \,\mathrm{d}s + \xi \int_t^{t+\Delta} \sqrt{v_s}\,\mathrm{d}W_s^2,$$

hence

$$\int_t^{t+\Delta} \sqrt{v_s}\,\mathrm{d}W_s^2 = \frac{1}{\xi}\Big(v_{t+\Delta} - v_t - \kappa\theta\Delta + \kappa\int_t^{t+\Delta} v_s \,\mathrm{d}s\Big).$$

Substituting, we obtain an *exact* identity:

$$X_{t+\Delta} = X_t + r\Delta + \frac{\rho}{\xi}(v_{t+\Delta} - v_t - \kappa\theta\Delta) + \Big(\frac{\kappa\rho}{\xi} - \frac{1}{2}\Big)\int_t^{t+\Delta} v_s \,\mathrm{d}s + \sqrt{1-\rho^2}\int_t^{t+\Delta} \sqrt{v_s}\,\mathrm{d}W_s^\perp. \quad (38)$$

All the correlation between the price and the variance now goes through the increment $v_{t+\Delta} - v_t$, which has already been simulated. The last term involves a Brownian motion independent of the variance path: conditionally on the latter, it is Gaussian, centred, with variance $(1-\rho^2)\int_t^{t+\Delta} v_s \,\mathrm{d}s$ (*Itô isometry*).

The only approximation. It remains to approximate the integral of the variance. We use a two-point quadrature rule,

$$\int_t^{t+\Delta} v_s \,\mathrm{d}s \approx (\gamma_1 v_t + \gamma_2 v_{t+\Delta})\Delta, \qquad \gamma_1 + \gamma_2 = 1,$$

with $\gamma_1 = \gamma_2 = \frac{1}{2}$ (trapezoidal rule). Substituting into (38), we obtain Andersen's scheme:

$$X_{t+\Delta} = X_t + r\Delta + K_0 + K_1 v_t + K_2 v_{t+\Delta} + \sqrt{K_3 v_t + K_4 v_{t+\Delta}}\, Z_S, \tag{39}$$

where $Z_S \sim \mathcal{N}(0,1)$ is *independent* of the variance, and

$$K_0 = -\frac{\rho\kappa\theta}{\xi}\Delta, \qquad K_1 = \gamma_1\Delta\Big(\frac{\kappa\rho}{\xi} - \frac{1}{2}\Big) - \frac{\rho}{\xi}, \qquad K_2 = \gamma_2\Delta\Big(\frac{\kappa\rho}{\xi} - \frac{1}{2}\Big) + \frac{\rho}{\xi},$$
$$K_3 = \gamma_1\Delta(1-\rho^2), \qquad K_4 = \gamma_2\Delta(1-\rho^2).$$

Comparison with Euler. The Euler scheme (32) freezes the variance at its value v_t at the start of the step, and introduces the correlation through a Gaussian Z_S correlated with Z_v. Scheme (39), on the contrary, uses both endpoints v_t and $v_{t+\Delta}$, and the correlation is treated exactly thanks to identity (38). The only error comes from the trapezoidal rule.

7.8 Martingale correction

The problem. Under $\mathbb{Q}$, the discounted price is a martingale: $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$. Discretised schemes do not satisfy this property exactly: scheme (39) satisfies $\mathbb{E}[e^{X_{t+\Delta}} \mid \mathcal{F}_t] = e^{X_t + r\Delta}$ only up to $O(\Delta)$. Andersen proposes an analytical correction of K_0; in practice, an empirical correction is often used.

Empirical correction of Duan and Simonato (1998). Once the N paths have been simulated, each $S_T^{(i)}$ is replaced by

$$\widetilde{S}_T^{(i)} = S_T^{(i)} \frac{S_0 e^{rT}}{\overline{S_T}}, \qquad \overline{S_T} = \frac{1}{N}\sum_{j=1}^{N} S_T^{(j)},$$

which imposes exactly $\overline{\widetilde{S}_T} = S_0 e^{rT}$. This correction reduces the variance and corrects part of the bias on the drift. It has two drawbacks: the $\widetilde{S}_T^{(i)}$ are no longer independent, so that the interval (29) is no longer rigorously justified; and it can *mask* the bias of a scheme, which distorts their comparison. For this reason, **all the bias tests in this section are carried out without martingale correction**.

To reduce the variance without losing the independence of the paths, we prefer the methods of Section 7.10.

7.9 Numerical results of standard Monte Carlo

Case	κ	θ	ξ	ρ	v_0	$2\kappa\theta - \xi^2$	d	Lewis price
Feller satisfied	3	0.04	0.3	-0.6	0.04	$+0.150$	5.33	9.308051
Feller violated	1	0.04	1.2	-0.7	0.04	-1.360	0.11	7.022250

Table 3: Parameters of the Monte Carlo tests. In both cases, $S_0 = 100$, $K = 100$, $T = 1$ and $r = 0.03$.

7.9.1 Test 1: law of the variance over one step

We simulate a single step of size Δ from v_0, in the case where Feller is violated, with 400,000 draws. Each scheme is compared with a sample from the exact law (noncentral chi-squared). The Wasserstein distance $W_1(\mu,\nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)|\,\mathrm{d}x$ is the relevant indicator here: by Kantorovich duality, it bounds the gap $|\mathbb{E}_\mu[f] - \mathbb{E}_\nu[f]|$ for every 1-Lipschitz function f, hence the error on a price. For Euler, the distance is computed after truncation $v \mapsto v^+$.

		Distance W_1		$\mathbb{Q}(v_\Delta \le 10^{-3})$			
Δ	ψ	QE	Euler	exact	QE	Euler	Exact / QE std. dev.
0.50	11.38	0.0050	0.0646	0.691	0.840	0.408	0.1348 / 0.1349
0.10	3.26	0.0006	0.0234	0.483	0.538	0.303	0.0719 / 0.0723
0.02	0.71	0.0005	0.0053	0.067	0.038	0.125	0.0336 / 0.0337

Table 4: Law of v_Δ given v_0, Feller violated: comparison with the exact law.

Table 4 and Figure 2 confirm the theoretical analysis.



Figure 2: Law of v_Δ given v_0 for $\Delta = 0.1$, Feller violated. The exact law concentrates nearly half of its mass in the immediate vicinity of 0; QE reproduces this shape, whereas Euler produces a Gaussian that spills widely below 0.

- **Euler** is structurally unsuited: for $\Delta = 0.1$, about 30 % of the draws are negative, and its distance W_1 to the exact law is 10 to 40 times larger than that of QE.
- **QE** reproduces the exact law, including its standard deviation, to within 1 % at most, as guaranteed by moment matching. For $\Delta = 0.5$ and 0.1 ($\psi > \psi_c$), the exponential branch is at work: its mass at 0 approximates the concentration of the exact law near 0. For $\Delta = 0.02$ ($\psi < \psi_c$), it is the quadratic branch.

7.9.2 Test 2: discretisation bias

We simulate $N = 500{,}000$ paths for different numbers of steps, without martingale correction. Errors are expressed in basis points of the Lewis price, with the half-width of the confidence interval (29): a gap smaller than this half-width is not statistically significant.

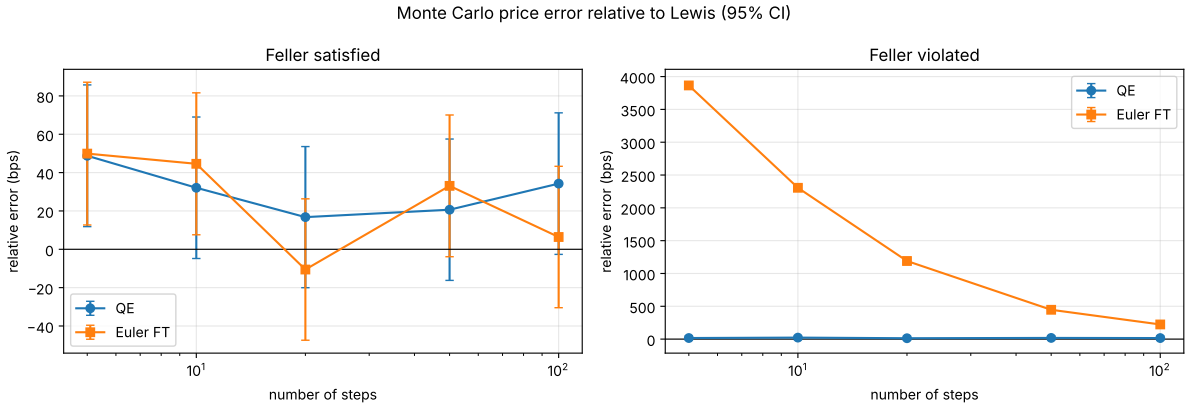


Figure 3: Relative error of the Monte Carlo price as a function of the number of steps (95 % CI).

	Feller satisfied		Feller violated				
n	QE (bps)	Euler (bps)	QE (bps)	Euler (bps)	$v < 0$ (Euler)	bias $\mathbb{E}[v_T]$ QE	bias $\mathbb{E}[v_T]$ Euler
5	$+48.8 \pm 37.0$	$+49.9 \pm 37.2$	$+16.5 \pm 30.6$	$+3865 \pm 49$	43.3 %	+0.0002	+0.0321
10	$+32.1 \pm 36.9$	$+44.6 \pm 37.0$	$+22.8 \pm 30.4$	$+2306 \pm 41$	53.5 %	+0.0004	+0.0148
20	$+16.8 \pm 36.8$	-10.6 ± 36.9	$+13.4 \pm 30.4$	$+1190 \pm 35$	58.0 %	−0.0000	+0.0063
50	$+20.6 \pm 36.9$	$+33.1 \pm 37.0$	$+18.0 \pm 30.5$	$+447 \pm 32$	57.9 %	+0.0004	+0.0017
100	$+34.3 \pm 36.9$	$+6.4 \pm 36.9$	$+15.9 \pm 30.4$	$+223 \pm 31$	55.7 %	+0.0003	+0.0007

Table 5: Error of the Monte Carlo price relative to Lewis (95 % CI), $N = 500{,}000$. In both cases, $\mathbb{E}[v_T] = 0.04$.

Feller satisfied. The two schemes are indistinguishable. Almost all errors are smaller than the half-width of the confidence interval: at this level of precision, no bias is measurable. Euler's truncation almost never comes into play (less than 2 % of steps from $n = 10$ onwards).

Feller violated. The behaviour of the two schemes diverges completely.

- **Euler** massively overestimates the price: +3865 bps (i.e. +39 %) for 5 steps, and still +223 bps for 100 steps. More than half of the steps are truncated. Truncation replaces a negative variance by 0, which systematically biases the variance upwards: $\mathbb{E}[v_T]$ is overestimated by 80 % for 5 steps.
- **QE** has no significant bias, from 5 steps onwards: all errors are smaller than the confidence interval. The bias on $\mathbb{E}[v_T]$ is zero up to statistical fluctuations, as predicted by the exact moment matching.

7.9.3 Test 3: statistical error

With the scheme fixed (QE, 50 steps, Feller violated), we increase the number of paths from 10^3 to 10^6. Table 6 gives the estimated price, the half-width of the 95 % confidence interval, $1.96\,\widehat{\sigma}_N/\sqrt{N}$, and the realised error relative to the Lewis price. The half-width is plotted on a log-log scale in Figure 4.

N	MC price	95 % CI	$\lvert\widehat{C}_N - C\rvert$	In the CI?
1,000	6.8424	±0.4419	0.1798	yes
3,000	6.8025	±0.2711	0.2197	yes
10,000	6.9807	±0.1496	0.0416	yes
30,000	7.0728	±0.0891	0.0506	yes
100,000	7.0777	±0.0479	0.0555	no
300,000	7.0300	±0.0275	0.0078	yes
1,000,000	7.0195	±0.0150	0.0028	yes

Table 6: Monte Carlo price, half-width of the 95 % confidence interval and realised error as a function of the number of paths (QE, 50 steps, Feller violated). Lewis price: 7.0223.

The measured slope is -0.492, in line with the theoretical rate $N^{-1/2}$ given by the central limit theorem: dividing the interval by 10 requires 100 times more paths.

The realised error, available here thanks to the Lewis price, serves as a check: it should fall within the interval in about 95 % of cases. This is the case for 6 values of N out of 7. The only exception, marginal (0.0555 versus 0.0479 for $N = 10^5$), is compatible with a 95 % interval, which leads one to expect one exit in twenty. This realised error is random and does not decrease regularly: this is why the rate of convergence is read on the half-width of the interval, and not on the error itself.

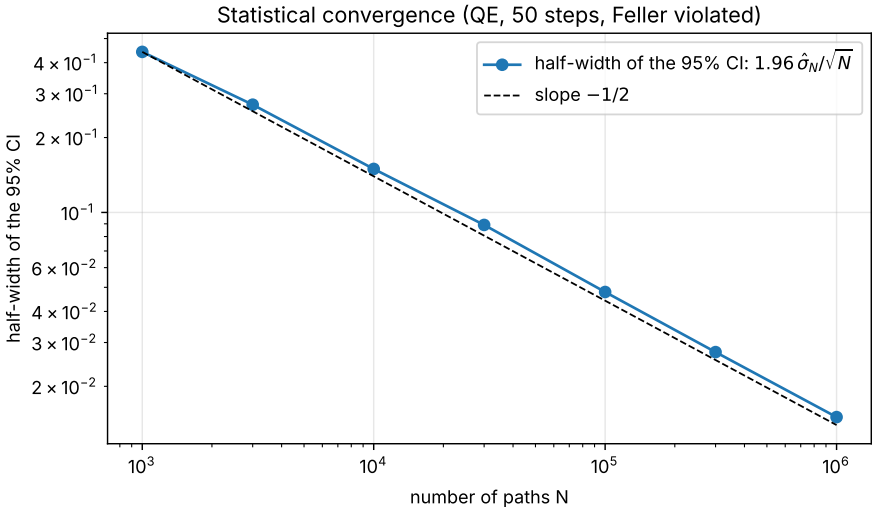


Figure 4: Half-width of the 95 % confidence interval as a function of the number of paths N (QE, 50 steps, Feller violated). The dashed line has slope $-1/2$.

To reach a half-width of 0.015, i.e. about 20 basis points of the price, 10^6 paths are needed. It is this cost that variance reduction seeks to lower.

7.9.4 Test 4: computation time

For 200,000 paths and 100 steps, QE takes about 1.9 s and Euler about 1.0 s: QE is therefore 1.8 to 2.2 times more expensive per step. This extra cost comes from the computation of m, s^2, ψ and of the two branches. It is largely offset by the number of steps required: when Feller is violated, QE reaches with 5 steps an accuracy that Euler would only reach with several hundred steps. For the same accuracy, QE is therefore of the order of 50 to 100 times faster than Euler in this regime. For comparison, one price with the Lewis formula takes about 2.5 ms.

7.9.5 Test 5: cross-check on a full smile

Finally, we compare the implied volatility smile obtained by Monte Carlo (QE, 100 steps, 10^6 paths, a single simulation reused for all strikes) with that of the Lewis formula, in the case where Feller is violated.

The two smiles overlap over the whole range of strikes, from 70 to 140; the maximum implied volatility gap is 3.5 bps. This test is the most complete: it simultaneously validates the simulation scheme, the Lewis formula and the consistency of the whole on in-, at- and out-of-the-money options.

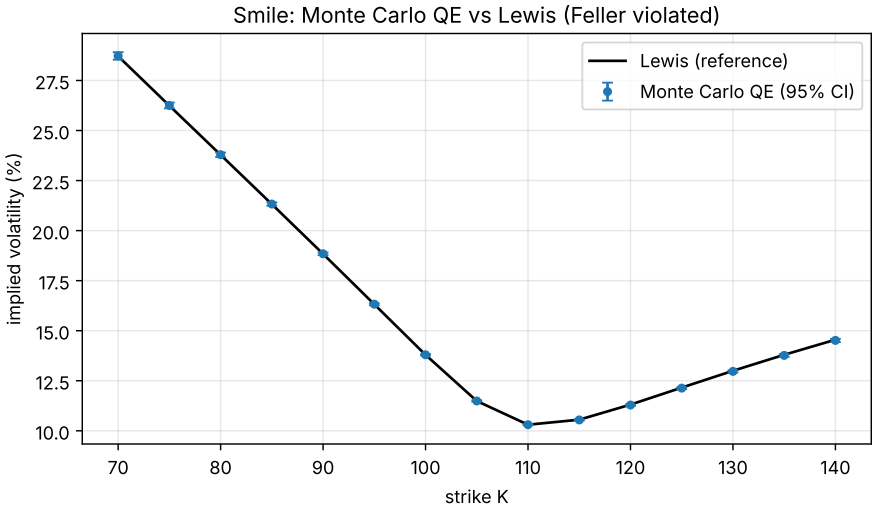


Figure 5: Implied volatility smile: Monte Carlo QE (95 % CI) versus Lewis, Feller violated.

7.10 Variance reduction

7.10.1 Principle and plan

With the scheme fixed, the statistical error is $1.96\,\sigma/\sqrt{N}$: halving the error costs 4 times more paths. A variance reduction method acts on the other factor: it replaces the simulated variable $Y = e^{-rT}(S_T - K)^+$ by a variable $\widetilde{Y}$ *with the same expectation*, but with lower variance. The right comparison criterion is not the variance alone: for the same accuracy, the required time is proportional to $\mathrm{Var}(\widetilde{Y}) \times t(\widetilde{Y})$, where t is the cost of one path. We therefore define the *efficiency gain*

$$\mathcal{G} = \frac{\mathrm{Var}(Y)\; t(Y)}{\mathrm{Var}(\widetilde{Y})\; t(\widetilde{Y})}. \tag{40}$$

We study three methods, all built on the QE scheme:

1. **conditional Monte Carlo**;
2. a **control variate**;
3. **conditional Monte Carlo combined with a control variate**.

For each, we explain the idea, then prove why the variance decreases and in which situations the method is effective. The numerical results are gathered at the end.

7.10.2 Method 1: conditional Monte Carlo

Conditionally on the variance path, the log-price is Gaussian: the option price is then given by the Black-Scholes formula (Willard, 1997; Romano and Touzi, 1997). We therefore simulate only the variance, and replace the payoff by this conditional price. We denote by $\mathcal{G}$ the σ-algebra generated by W^2, which contains all the information on the variance path, and $I_T = \int_0^T v_s \, \mathrm{d}s$.

Proposition 5 (Conditional price)**.**

$$\mathbb{E}^{\mathbb{Q}}\Big[e^{-rT}(S_T - K)^+ \,\Big|\, \mathcal{G}\Big] = C_{\mathrm{BS}}(S_{\mathrm{eff}}, K, T, r, \sigma_{\mathrm{eff}}), \tag{41}$$

with

$$S_{\mathrm{eff}} = S_0 \exp\Big(\frac{\rho}{\xi}(v_T - v_0 - \kappa\theta T) + \frac{\kappa\rho}{\xi} I_T - \frac{\rho^2}{2} I_T\Big), \qquad \sigma^2_{\mathrm{eff}} = \frac{(1-\rho^2)\, I_T}{T}. \tag{42}$$

Proof. We write $W^1 = \rho W^2 + \sqrt{1-\rho^2}\, W^\perp$, with $W^\perp$ a Brownian motion independent of W^2. The exact identity (38), applied on $[0, T]$, gives

$$X_T = \mu + \sqrt{1-\rho^2} \int_0^T \sqrt{v_s}\, \mathrm{d}W_s^\perp, \qquad \mu = X_0 + rT + \frac{\rho}{\xi}(v_T - v_0 - \kappa\theta T) + \Big(\frac{\kappa\rho}{\xi} - \frac{1}{2}\Big) I_T.$$

The variance is a function of W^2 alone, so μ and I_T are known given $\mathcal{G}$. Given $\mathcal{G}$, the integrand $\sqrt{v_s}$ is deterministic and $W^\perp$ remains a Brownian motion: the integral is Gaussian, centred, with variance I_T. Thus

$$X_T \mid \mathcal{G} \;\sim\; \mathcal{N}(\mu,\; (1-\rho^2)\, I_T).$$

Now, in the Black-Scholes model with spot S and volatility σ, $\ln S_T \sim \mathcal{N}\big(\ln S + (r - \tfrac{1}{2}\sigma^2)T,\; \sigma^2 T\big)$. The two laws coincide for $\sigma^2_{\mathrm{eff}} T = (1-\rho^2) I_T$ and $\ln S_{\mathrm{eff}} = \mu - rT + \tfrac{1}{2}(1-\rho^2) I_T$, which gives (42). □

Variance reduction. The conditional estimator $Y^{\mathrm{cond}} = \mathbb{E}[Y \mid \mathcal{G}]$ has the same expectation as Y (tower property), and the law of total variance gives

$$\mathrm{Var}(Y) = \mathrm{Var}(Y^{\mathrm{cond}}) + \mathbb{E}\big[\,\mathrm{Var}(Y \mid \mathcal{G})\big] \;\geq\; \mathrm{Var}(Y^{\mathrm{cond}}). \tag{43}$$

We remove exactly the part of the variance due to $W^\perp$, which is strictly positive as soon as $|\rho| < 1$.

7.10.3 Method 2: control variate

We use a variable X, simulated on the same paths as Y, correlated with Y and with known expectation, to correct Y by the deviation observed on X:

$$Y_\beta = Y - \beta(X - \mathbb{E}[X]), \qquad \mathbb{E}[Y_\beta] = \mathbb{E}[Y] \text{ for all } \beta.$$

Proposition 6 (Optimal control variate)**.** *The variance of Y_β is minimal for $\beta^* = \mathrm{Cov}(X, Y)/\,\mathrm{Var}(X)$, and*

$$\mathrm{Var}(Y_{\beta^*}) = \mathrm{Var}(Y)\,(1 - \varrho^2), \qquad \varrho = \mathrm{corr}(X, Y).$$

Proof. $\mathrm{Var}(Y_\beta) = \mathrm{Var}(Y) - 2\beta\,\mathrm{Cov}(X, Y) + \beta^2\,\mathrm{Var}(X)$ is a quadratic in β, minimal at β^*; substituting, we obtain $\mathrm{Var}(Y) - \mathrm{Cov}(X, Y)^2/\,\mathrm{Var}(X) = \mathrm{Var}(Y)(1 - \varrho^2)$. □

Estimation of β^*. The optimal coefficient $\beta^* = \mathrm{Cov}(X,Y)/\mathrm{Var}(X)$ involves an unknown covariance and variance. They are replaced by their empirical estimators, computed on the same N paths, which gives $\hat{\beta}$, the slope of the linear regression of Y on X. The estimator becomes

$$\widehat{C}^{\mathrm{VC}} = \overline{Y} - \hat{\beta}\,(\overline{X} - \mathbb{E}[X]),$$

where $\overline{Y}$ and $\overline{X}$ are the empirical means. If β were a fixed constant, the second term would have zero expectation. But $\hat{\beta}$ is random, and it is computed from the same draws as $\overline{X}$: the two are not independent, and the expectation of their product is not zero. Since $\hat{\beta} - \beta^*$ and $\overline{X} - \mathbb{E}[X]$ are each of order $1/\sqrt{N}$, their product, hence the bias, is of order $1/N$. It is negligible: the statistical error is of order $1/\sqrt{N}$.

The chosen control: $X = e^{-rT}S_T$. This control combines the three desired qualities:

- its expectation is known exactly, $\mathbb{E}[e^{-rT}S_T] = S_0$, by the martingale property (8);
- it is strongly correlated with the payoff, which is an increasing function of S_T, and almost linear in the money;
- it costs nothing: S_T is already simulated.

A precaution: the expectation of the simulated control. The estimator uses the simulated variable $\hat{X} = e^{-rT}\hat{S}_T$, and S_0 is written in place of its expectation. Now, the QE scheme satisfies the martingale property only up to $O(\Delta)$: $\mathbb{E}[\hat{X}] = S_0 + \varepsilon_\Delta$. This choice therefore adds a bias $-\beta\,\varepsilon_\Delta$, which does not decrease with N, but which is of the same order as the discretisation bias already present in the scheme.

Chosen solution. We keep S_0 as the expectation of the control, and check that ε_Δ is negligible. With 10^6 paths and 50 steps, we measure $\mathbb{E}[\hat{X}] - S_0 = +0.018 \pm 0.038$ when Feller is satisfied, and $+0.014 \pm 0.031$ when it is violated: the gap is indistinguishable from 0. The comparison with the Lewis price over all strikes (Table 8) confirms the absence of measurable bias. If greater accuracy were needed, Andersen's analytical martingale correction, applied to the constant K_0 of the scheme, would make the equality $\mathbb{E}[\hat{X}] = S_0$ exact.

7.10.4 Method 3: conditional Monte Carlo and control variate

Why methods 1 and 2 are effective in opposite regions. For each method, we look at which part of the variance of $Y = e^{-rT}(S_T - K)^+$ it removes. The table below gives the gains measured at the two ends of the strike range (Figure 6).

	Feller satisfied		Feller violated	
	$K = 70$ (in the money)	$K = 140$ (out)	$K = 70$	$K = 140$
Method 1: conditional	×3.0	×80	×3.2	×40
Method 2: control variate	×79	×1.1	×10	×1.1

Method 2: it removes the linear part of the payoff. Path by path,

$$e^{-rT}(S_T - K)^+ = \underbrace{e^{-rT}S_T - Ke^{-rT}}_{\text{linear function of the control}} + \underbrace{e^{-rT}(K - S_T)^+}_{\text{remainder: put payoff}}.$$

The control $e^{-rT}S_T$ removes exactly the first term (with $\beta = 1$). It can do nothing about the second, which is not a linear function of S_T. The residual variance is therefore at most that of the put payoff. Everything depends on the relative size of this remainder:

- *in the money* (small K), S_T exceeds K on almost all paths: the put is almost always zero, the remainder is tiny, and the gain is very large ($\times 79$);
- *out of the money* (large K), it is the opposite: the put is almost always active, and the remainder contains almost all of the variance of the call. The gain tends to 1 ($\times 1.1$).

Method 1: it removes the randomness of $W^\perp$. By (43), conditioning removes exactly $\mathbb{E}[\mathrm{Var}(Y \mid \mathcal{G})]$, the part of the variance due to the Brownian motion $W^\perp$. Its effectiveness depends on the size of this part:

- *in the money*, the payoff is almost linear in S_T, so its variance is essentially that of the noise of $\ln S_T$. By the proof of Proposition 5, this noise is the sum of $\rho \int_0^T \sqrt{v_s}\, \mathrm{d}W_s^2$ and $\sqrt{1-\rho^2} \int_0^T \sqrt{v_s}\, \mathrm{d}W_s^\perp$. These two terms are uncorrelated, with variances $\rho^2\, \mathbb{E}[I_T]$ and $(1-\rho^2)\, \mathbb{E}[I_T]$ by the Itô isometry. Conditioning only removes the second: a fraction ρ^2 of the variance remains, and the gain is about $1/\rho^2 = 2.8$ for $\rho = -0.6$. This is indeed what we measure ($\times 3$);
- *out of the money*, the payoff is zero on most paths and large on a few: its variance is huge compared with its expectation. Most of this variance comes from the randomness that decides whether S_T exceeds K, that is, from the noise of $W^\perp$, which conditioning removes entirely. The conditional price, on the other hand, is strictly positive on all paths, and only varies with the variance path. The gain becomes very large ($\times 80$).

The two methods are therefore complementary: method 2 removes the linear part of the payoff, which dominates in the money; method 1 removes the noise of $W^\perp$, which dominates out of the money. To be effective everywhere, we combine them, by applying a control variate to the conditional estimator.

The control: S_{eff}. For the conditional estimator, the natural control is S_{eff}, which plays for $Y^{\text{cond}} = C_{\text{BS}}(S_{\text{eff}}, K, T, r, \sigma_{\text{eff}})$ the role that S_T plays for the payoff. The estimator is

$$\widehat{C}^{(3)} = \frac{1}{N} \sum_{i=1}^{N} \Big(Y_i^{\text{cond}} - \beta (S_{\text{eff},i} - S_0) \Big),$$

with β estimated by regression, as for method 2. The expectation of this control must be known.

Lemma 1 (S_{eff} is the conditional expectation of the control of method 2)**.**

$$\mathbb{E}\big[e^{-rT} S_T \mid \mathcal{G}\big] = S_{\text{eff}}, \qquad \textit{hence} \qquad \mathbb{E}[S_{\text{eff}}] = S_0.$$

Proof. By the proof of Proposition 5, given $\mathcal{G}$, X_T is Gaussian with mean μ and variance $(1-\rho^2) I_T$, with $\mu + \frac{1}{2}(1-\rho^2) I_T = \ln S_{\text{eff}} + rT$. For a Gaussian G with mean m and variance s^2, $\mathbb{E}[e^G] = e^{m + s^2/2}$; hence $\mathbb{E}[e^{X_T} \mid \mathcal{G}] = S_{\text{eff}}\, e^{rT}$. The tower property and the martingale property (8) then give $\mathbb{E}[S_{\text{eff}}] = \mathbb{E}[e^{-rT} S_T] = S_0$. □

As for method 2, this equality holds under the simulated scheme only up to discretisation; the same solution applies: we use S_0, and check the absence of bias against the Lewis price.

Proposition 7 (The combination beats both methods)**.** *The variance of method 3, with the optimal coefficient, is lower than both that of method 1 (conditional alone) and that of method 2 (control $e^{-rT} S_T$ alone).*

Proof. Against method 1. The choice $\beta = 0$ gives back exactly the conditional estimator. The optimal coefficient does at least as well as this particular choice.

Against method 2. We denote by $Z = Y - \beta_2(e^{-rT}S_T - S_0)$ the estimator of method 2, with its optimal coefficient β_2. We condition on $\mathcal{G}$. By definition of Y^{cond} and by Lemma 1,

$$\mathbb{E}[Z \mid \mathcal{G}] = Y^{\text{cond}} - \beta_2(S_{\text{eff}} - S_0) :$$

this is the estimator of method 3, with the coefficient $\beta = \beta_2$. Now conditioning always decreases the variance, by (43): $\text{Var}(Z) \geq \text{Var}\left(\mathbb{E}[Z \mid \mathcal{G}]\right)$. Finally, the optimal coefficient of method 3 does at least as well as $\beta = \beta_2$. □

7.10.5 Results: comparison at the money

We compare the three methods with standard Monte Carlo, with $N = 200{,}000$ paths, 50 time steps and $K = 100$. The time is that of the simulation; computing the estimators (means, regression, Black-Scholes formula) takes a few milliseconds.

We plot the variance gain of each method for K from 70 to 140, in both cases, reusing the same simulation for all strikes.

Method	Feller satisfied (Lewis: 9.3081)				Feller violated (Lewis: 7.0223)			
	Price	95 % CI	Var. gain	Effic. gain	Price	95 % CI	Var. gain	Effic. gain
Standard	9.2998	±0.0542	1	1	7.0073	±0.0337	1	1
1. Conditional	9.3030	±0.0251	4.7	6.0	7.0243	±0.0071	22.6	28.3
2. Control variate	9.2906	±0.0254	4.5	4.5	7.0181	±0.0235	2.0	2.0
3. Cond. + control	9.3016	±0.0082	43.5	55.9	7.0196	±0.0045	56.1	70.2

Table 7: Variance reduction at the money ($K = 100$, 50 steps, 200,000 paths). Gains relative to standard Monte Carlo, for the same number of paths (variance) and for the same accuracy (efficiency, defined in (40)).

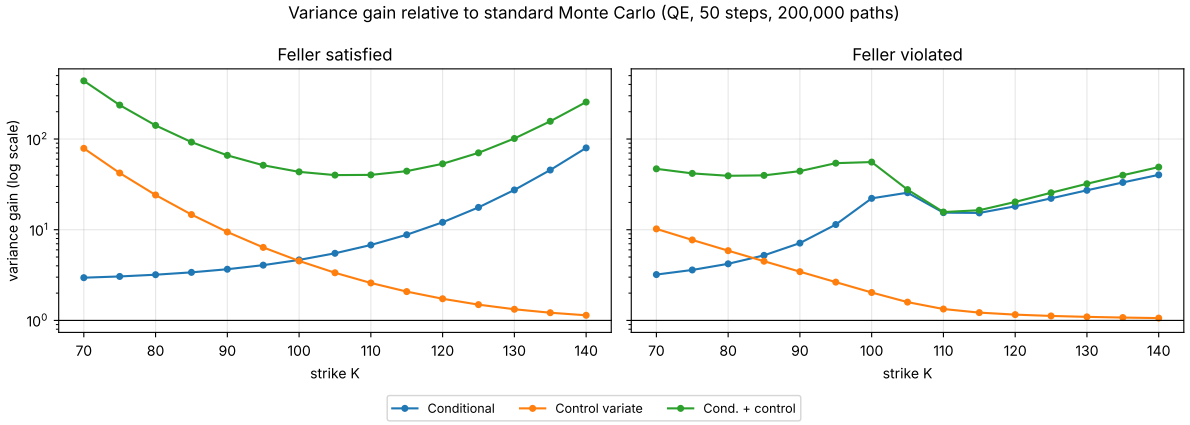


Figure 6: Variance gain relative to standard Monte Carlo, as a function of the strike (QE, 50 steps, 200,000 paths).

Method	Feller satisfied			Feller violated		
	Min gain	Max gain	Max gap	Min gain	Max gain	Max gap
1. Conditional	3.0 ($K = 70$)	80 ($K = 140$)	0.38	3.2 ($K = 70$)	40 ($K = 140$)	0.46
2. Control variate	1.1 ($K = 140$)	79 ($K = 70$)	0.37	1.1 ($K = 140$)	10 ($K = 70$)	0.62
3. Cond. + control	40	438 ($K = 70$)	0.26	15.7	56	0.48

Table 8: Minimum and maximum variance gain over the 15 strikes from 70 to 140, and maximum gap to the Lewis price, expressed in confidence-interval half-widths.

No bias. Over the 15 strikes and for the three methods, the gap to the Lewis price remains below 0.62 confidence-interval half-widths. The possible bias of the control variates, whose expectation is exact only up to discretisation, is therefore not measurable.

Summary. For a European option, the combination of conditioning and the control variate S_{eff} is the method to retain: it is the most effective at the money, and the only one that remains effective over the whole strike range and in both Feller regimes, at a cost lower than that of standard Monte Carlo.

Limitation. Conditioning assumes that the payoff depends only on S_T. For a barrier or Asian option, the full path of the underlying is needed, and the method no longer applies as such. The control variate remains available; one can also use as a control the European call simulated on the same paths, whose exact price is given by the Lewis formula.

8 Calibration

8.1 The calibration problem

The previous sections compute prices from known parameters. Calibration solves the inverse problem: finding the parameters $\Theta = (\kappa, \theta, \xi, \rho, v_0)$ for which the model best reproduces the prices and the smile observed on the market. We have N quoted call options, with strikes K_i, maturities T_i and market prices C_i^{mkt}, $i = 1, \dots, N$. We denote by $C_i(\Theta)$ the corresponding Heston price, computed with the Lewis formula (28). We seek

$$\widehat{\Theta} = \arg\min_{\Theta \in \mathcal{B}} \ \frac{1}{2} \sum_{i=1}^{N} r_i(\Theta)^2, \tag{44}$$

where $r_i(\Theta)$ is a residual measuring the gap between the model and the market for option i (Section 8.2), and $\mathcal{B}$ a box of bounds:

$$\kappa \in [10^{-3}, 10], \quad \theta \in [10^{-4}, 1], \quad \xi \in [10^{-3}, 2], \quad \rho \in [-0.999,\ 0.999], \quad v_0 \in [10^{-4}, 1].$$

These bounds guarantee that the characteristic function remains well defined, and that no parameter takes an economically absurd value. A calibrated parameter that sticks to one of them is a warning sign: the problem is then ill-posed (Section 8.5).

The Feller condition (4) is deliberately *not* imposed. The pricing formulas remain valid when it is violated, and the QE scheme simulates this regime correctly (Section 7). Imposing it would amount to artificially constraining the vol of vol, at the expense of the fit of the smile.

8.2 Choice of the loss function

The loss function determines how the gap between the model and the market is measured, and hence the weight that each option receives in the calibration. We compare three choices of residuals $r_i(\Theta)$. The key point is the following: the errors of a model are naturally measured in *implied volatility*, the unit in which the market reasons, whereas the data are *prices*.

(a) Price loss.

$$r_i^{\text{price}}(\Theta) = C_i(\Theta) - C_i^{\text{mkt}}.$$

This is the simplest choice: prices are compared directly. Its drawback is that the same model error does not produce the same price gap on all options. If the model is wrong by $\Delta\sigma_i$ on the implied volatility of option i, the price gap is, to first order,

$$r_i^{\text{price}} \approx \mathcal{V}_i \, \Delta\sigma_i, \qquad \text{hence} \qquad \sum_i \left(r_i^{\text{price}}\right)^2 \approx \sum_i \mathcal{V}_i^2 \, \Delta\sigma_i^2,$$

where $\mathcal{V}_i$ is the vega of the option, that is, the sensitivity of its price to volatility. The price loss therefore weights each option by the *square of its vega*. This weight is very uneven: with a volatility of 20 %, an error of 1 volatility point changes the price of an at-the-money option with a 2-year maturity by 0.53, but that of an option with maturity 0.1 year and strike 120 by only 0.0025. The price loss thus favours at-the-money and long-maturity options, and neglects short-maturity options far from the money. Yet it is precisely these wings of the smile that shape it: its asymmetry, set by ρ, and its curvature, set by ξ. It therefore fits the level of at-the-money volatility well, but poorly constrains the shape of the smile.

(b) Vega-weighted price loss.

$$r_i^{\text{vega}}(\Theta) = \frac{C_i(\Theta) - C_i^{\text{mkt}}}{\mathcal{V}_i}, \qquad \mathcal{V}_i = \frac{\partial C_{\text{BS}}}{\partial \sigma}(K_i, T_i, \sigma_i^{\text{mkt}}),$$

where σ_i^{mkt} is the market implied volatility. Dividing each price gap by the vega exactly cancels the uneven weighting of loss (a): by the previous expansion,

$$r_i^{\text{vega}} \approx \Delta\sigma_i = \sigma_i(\Theta) - \sigma_i^{\text{mkt}},$$

where $\sigma_i(\Theta)$ is the implied volatility of the Heston price. This residual is therefore, to first order, an implied volatility gap, and all options are treated on an equal footing. Its interest is that it is obtained *without inverting* the Black-Scholes formula: the market vega $\mathcal{V}_i$ is computed once and for all, before the optimisation, and each evaluation only requires the model prices. Two limitations: the first-order approximation degrades when the volatility gap is large; and for an option with a very small vega, the division strongly amplifies any small quotation error in price. Options with near-zero vega are therefore discarded.

(c) Implied volatility loss.

$$r_i^{\text{IV}}(\Theta) = \sigma_i(\Theta) - \sigma_i^{\text{mkt}}.$$

This is the exact measure, without approximation, and the closest to market practice. But to obtain $\sigma_i(\Theta)$, the Black-Scholes formula must be inverted, with Brent's method, for each option and at each evaluation of the objective function. When the inversion fails, because the model price falls outside the no-arbitrage bounds, we fall back on residual (b).

The trade-off. The three losses are ranked according to two opposing criteria:

	(a) Price	(b) Price / vega	(c) Implied volatility
Error measure	in price, weighted by $\mathcal{V}_i^2$	in volatility, to first order	in volatility, exact
Cost of one evaluation	prices only	prices only	prices + N inversions

We therefore retain loss (b) by default, loss (c) serving as a check. Finally, to compare the three losses fairly, they are all evaluated on the same criterion, the implied volatility gap.

8.3 Optimisation algorithm & Acceleration: a vectorised Lewis pricer

Problem (44) is a bound-constrained nonlinear least-squares problem. We solve it with the *Trust Region Reflective* (TRF) algorithm, implemented in `scipy.optimize.least_squares`.
Each evaluation of the objective function requires the N model prices, and the optimiser requires several dozen of them. The cost of the pricer therefore determines the cost of the calibration. The first version of the project computed each price separately, with the adaptive quadrature `quad`.

Gauss-Legendre: In the Lewis formula (28), the strike only appears in the factor $\sqrt{K}\, e^{-\mathrm{i}u \ln K}$. The characteristic function $\varphi(u - \frac{\mathrm{i}}{2})$, which accounts for most of the cost, depends only on the maturity and the parameters. We therefore replace the adaptive quadrature by a Gauss-Legendre quadrature with n fixed nodes on $[0, u_{\max}]$:

$$\int_0^{u_{\max}} f(u)\,\mathrm{d}u \approx \sum_{j=1}^{n} w_j\, f(u_j),$$

where u_j and w_j are the Gauss-Legendre nodes and weights mapped onto $[0, u_{\max}]$. For a given maturity, we evaluate $\varphi(u_j - \frac{\mathrm{i}}{2})$ only once on the grid, then all strikes are obtained by a matrix-vector product:

$$C(K_l) = S_0 - \frac{\sqrt{K_l}\, e^{-rT}}{\pi} \sum_{j=1}^{n} w_j\, \frac{\mathrm{Re}\left[e^{-\mathrm{i}u_j \ln K_l}\, \varphi(u_j - \frac{\mathrm{i}}{2})\right]}{u_j^2 + \frac{1}{4}}.$$

The number of evaluations of φ thus goes from several hundred per option to n per maturity.

Accuracy and gain. With $n = 200$ and $u_{\max} = 200$, the gap with the adaptive quadrature is at most 7×10^{-11}, on three contrasting parameter sets, maturities from 0.05 to 2 years and strikes from 70 to 130. An evaluation of the whole test surface (54 options) takes about 1.4 ms, versus 170 to 230 ms with `quad`: a factor of 120 to 165.

8.4 Validation on a synthetic market

Why a synthetic market. On real data, the "true" parameters are unknown: one can measure the quality of the fit, but not the ability of the calibrator to recover the parameters. We therefore generate a surface from known parameters $\Theta^\star$, typical of an equity market, with a violated Feller condition as on AAPL (Table 9). The construction follows four steps:

1. computation of exact Heston prices with the Lewis formula, for 6 maturities (0.1; 0.25; 0.5; 1; 1.5; 2 years) and 9 strikes from 80 to 120, i.e. 54 options ($S_0 = 100$, $r = 0.03$);
2. conversion into implied volatilities;
3. addition of an independent Gaussian quotation noise, with a standard deviation of 15 bps of volatility;
4. conversion back into prices.

	κ	θ	ξ	ρ	v_0	$2\kappa\theta - \xi^2$
$\Theta^\star$	2	0.06	0.8	-0.65	0.045	-0.40

Table 9: Parameters of the synthetic market.

8.4.1 Error measures in implied volatility.

Consider a set of N options with observed market prices, indexed by $i \in \{1, \ldots, N\}$, with strike K_i and maturity T_i. We denote by σ_i^{mkt} the market implied volatility and by $\sigma_i^{\mathrm{mod}}(\Theta)$ the implied volatility obtained by inverting the Black-Scholes formula on the price $C^{\mathrm{mod}}(K_i, T_i; \Theta)$ given by the model with parameters Θ:

$$\sigma_i^{\mathrm{mod}}(\Theta) = \mathrm{BS}^{-1}\Big(C^{\mathrm{mod}}(K_i, T_i; \Theta);\, K_i, T_i\Big).$$

The mean absolute error in implied volatility (*Mean Absolute Error*) is defined by

$$\mathrm{MAE}_{\mathrm{IV}}(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\left|\sigma_i^{\mathrm{mkt}} - \sigma_i^{\mathrm{mod}}(\Theta)\right|,$$

and the root mean squared error (*Root Mean Squared Error*) by

$$\mathrm{RMSE}_{\mathrm{IV}}(\Theta) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\sigma_i^{\mathrm{mkt}} - \sigma_i^{\mathrm{mod}}(\Theta)\right)^2}.$$

These two indicators are expressed in volatility points, which makes it possible to compare on the same scale options with very different prices (ATM, OTM, short or long maturities), unlike an error measured directly in price. By the Cauchy-Schwarz inequality, we always have $\mathrm{MAE}_{\mathrm{IV}} \leq \mathrm{RMSE}_{\mathrm{IV}}$. The $\mathrm{RMSE}_{\mathrm{IV}}$ penalises large isolated errors more heavily: a marked gap between the two measures signals that the error is concentrated on a few options, typically the wings of the smile or the short maturities.

8.4.2 Test 1: recovery on exact prices

On noise-free prices, the calibrator (vega loss) recovers $\Theta^\star$ exactly, to the displayed precision, in 36 residual evaluations and 0.06 s, from the default starting point. This test validates the whole chain: pricer, residuals and optimiser. It also shows that a violated Feller condition poses no difficulty for the calibrator.

8.4.3 Test 2: comparison of loss functions

We then calibrate on the noisy market, with the three losses and the same starting point.

Loss	κ	θ	ξ	ρ	v_0	RMSE IV (bps)	Param. error	Evaluations	Time
$\Theta^\star$ (reference)	2.000	0.0600	0.800	−0.650	0.0450	13.6	–	–	–
Price	1.865	0.0617	0.780	−0.648	0.0442	14.8	2.86 %	60	0.13 s
Price / vega	2.026	0.0603	0.803	−0.649	0.0447	12.9	0.61 %	57	0.10 s
Implied volatility	2.024	0.0604	0.804	−0.649	0.0447	12.9	0.61 %	37	2.13 s

Table 10: Calibration on the noisy synthetic market (54 options, 15 bps noise). The parameter error is the mean relative gap to $\Theta^\star$. The $\Theta^\star$ row gives the error of the true parameters on the noisy market, that is, the noise level.

Default choice: the vega-weighted loss, as accurate as the implied volatility loss at a cost 20 times lower.

8.4.4 Test 3: quality of the fit

Figure 7 compares the market smiles with those of the calibrated model (vega loss), maturity by maturity. The model reproduces both the shape of the smile at each maturity, very asymmetric for short maturities, and its deformation with maturity, the skew flattening as T increases.

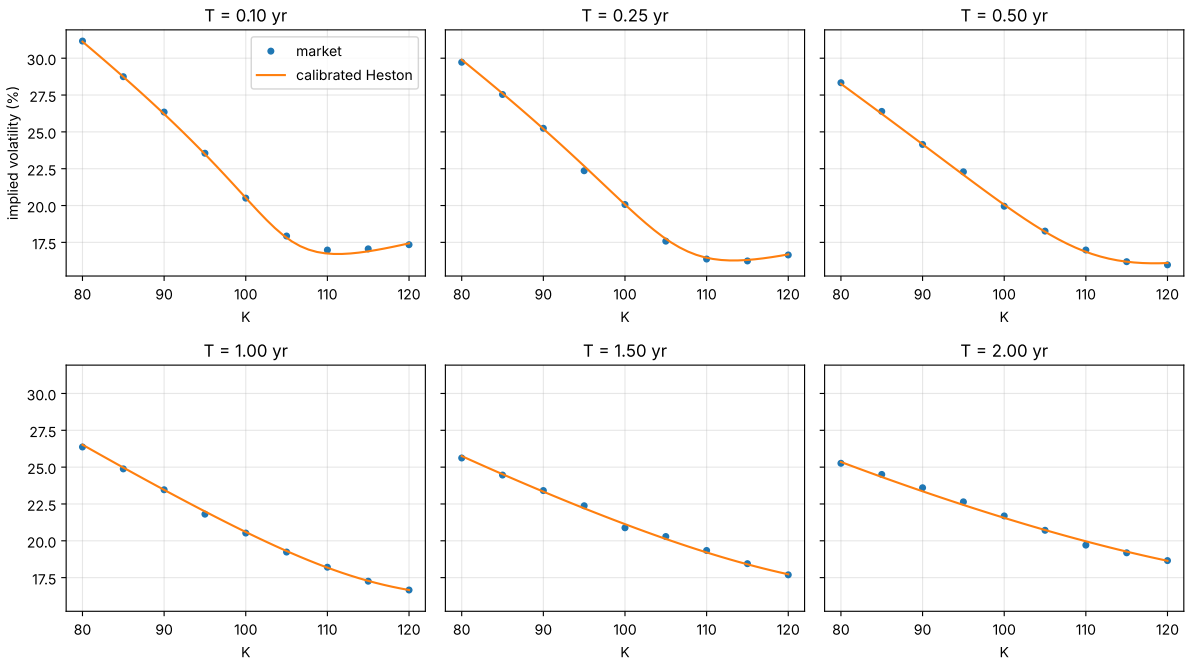


Figure 7: Implied volatility smiles of the noisy synthetic market (dots) and of the model calibrated with the vega loss (curves).

8.5 Identifiability of the parameters

A good fit does not guarantee good parameters: very different parameter sets can produce almost the same smile. The reason is that the smile of one maturity contains only three pieces of information, its level, its slope and its curvature, whereas the model has five parameters. The slope fixes ρ, but the level depends on κ, θ and v_0 only through the expected average variance over $[0, T]$: a high initial variance, quickly pulled back towards a low long-term level, gives the same average as a variance that rises slowly. Thus, calibrated on the single maturity $T = 0.5$, the model reproduces the smile perfectly with $\kappa = 5.15$, $\theta = 0.0001$ and $v_0 = 0.141$, instead of the true $\kappa = 2$, $\theta = 0.06$ and $v_0 = 0.045$. To separate these parameters, one must observe how the level evolves with maturity, hence calibrate on spread-out maturities, from a few weeks to more than a year: on the full surface, the calibration recovers the true parameters, and all starting points converge to the same solution. In practice, **a calibrated parameter set must be rejected** when a parameter sticks to one of its bounds (as $\theta = 0.0001$ above) or when the values are economically absurd, for example a near-zero long-term level θ or an initial variance v_0 very far from the square of the short-term implied volatility.

8.6 Computation time

Operation (54 options)	Time
Price evaluation, `quad` pricer	170 ms
Price evaluation, vectorised Gauss-Legendre pricer	1.4 ms
Evaluation of the implied volatility residuals	60 ms
Full calibration, vega loss	0.10 s
Full calibration, implied volatility loss	2.1 s

Table 11: Computation time of the calibration.

8.7 Application to real data

The notebook retrieves AAPL call options through the `yfinance` API (extraction on [DATE], spot $S_0 = $ [SPOT], risk-free rate $r = 0.04$, no dividend): quoted options (bid > 0 and ask > 0), mid price between bid and ask, a $\pm 15\,\%$ band around the spot, 8 maturities spread between 0.05 and 1.5 years. The quoted options are American and AAPL pays a dividend: these two effects are ignored here.

All the calibration functions apply without modification. With the first version of the calibrator, on 124 options, we had obtained:

Loss	κ	θ	ξ	ρ	v_0	$2\kappa\theta - \xi^2$	MAE IV	RMSE IV
Implied volatility	4.684	0.0986	1.161	-0.358	0.0559	-0.42	70 bps	112 bps
Price / vega	4.345	0.0987	1.027	-0.376	0.0551	-0.20	71 bps	114 bps

Table 12: Calibration on AAPL (124 options, 8 maturities), first version of the calibrator.

These results are interpreted in the light of the synthetic tests.

- **The two losses give almost identical fits** (70 and 71 bps), as on the synthetic market: the vega loss is the right default choice.
- **The Feller condition is violated**, as for $\Theta^\star$. Test 1 showed that this is not a defect of the calibrator: it is the market, through the shape of the smile, that pushes the model towards a high vol of vol.
- **κ and ξ differ noticeably from one loss to the other**, whereas θ, ρ and v_0 are stable. This is the identifiability hierarchy observed in Section 8.5: κ is the most fragile parameter, and slightly different losses are enough to move it.

9 Conclusion

This work built and validated a complete pricing chain in the Heston model, from the pricing formula to calibration.

Pricing. The affine structure of the model gives the characteristic function of the log-price in explicit form. The Gil-Pelaez and Lewis formulas, proved without any admitted result, deduce from it the price of the call; they agree to within 10^{-8} and converge to Black-Scholes as $\xi \to 0$.

Simulation. When the Feller condition is violated, the Euler scheme overestimates the price by 3865 bps with 5 steps. The QE scheme, which matches the first two moments of the exact CIR law, has no significant bias from 10 steps onwards and, for the same accuracy, is 50 to 100 times faster. Conditioning on the variance path, combined with the control variate S_{eff}, divides the variance by 43 to 56 at the money, and by at least 16 over the whole strike range.

Calibration. The loss divided by the vega is as accurate as the implied volatility loss, at a cost twenty times lower; with the vectorised Lewis pricer, a full calibration takes 0.10 s and recovers the true parameters to within $0.6\,\%$, at the level of the quotation noise. But a good fit does not guarantee good parameters: a single maturity constrains κ, θ and v_0 only through the average variance, and only a surface of spread-out maturities separates them.

References

[1] H. Albrecher, P. Mayer, W. Schoutens and J. Tistaert, *The little Heston trap*, Wilmott Magazine, 2007, pp. 83–92.

[2] L. Andersen, *Simple and efficient simulation of the Heston stochastic volatility model*, Journal of Computational Finance, 11(3), 2008, pp. 1–42.

[3] D. Bates, *Jumps and stochastic volatility: exchange rate processes implicit in Deutsche mark options*, Review of Financial Studies, 9(1), 1996, pp. 69–107.

[4] F. Black, *Studies of stock price volatility changes*, Proceedings of the American Statistical Association, Business and Economic Statistics Section, 1976, pp. 177–181.

[5] F. Black and M. Scholes, *The pricing of options and corporate liabilities*, Journal of Political Economy, 81(3), 1973, pp. 637–654.

[6] M. Broadie and Ö. Kaya, *Exact simulation of stochastic volatility and other affine jump diffusion processes*, Operations Research, 54(2), 2006, pp. 217–231.

[7] J. Cox, J. Ingersoll and S. Ross, *A theory of the term structure of interest rates*, Econometrica, 53(2), 1985, pp. 385–407.

[8] J.-C. Duan and J.-G. Simonato, *Empirical martingale simulation for asset prices*, Management Science, 44(9), 1998, pp. 1218–1233.

[9] D. Duffie, D. Filipović and W. Schachermayer, *Affine processes and applications in finance*, Annals of Applied Probability, 13(3), 2003, pp. 984–1053.

[10] W. Feller, *Two singular diffusion problems*, Annals of Mathematics, 54(1), 1951, pp. 173–182.

[11] J. Gil-Pelaez, *Note on the inversion theorem*, Biometrika, 38(3–4), 1951, pp. 481–482.

[12] S. Heston, *A closed-form solution for options with stochastic volatility with applications to bond and currency options*, Review of Financial Studies, 6(2), 1993, pp. 327–343.

[13] B. Horvath, A. Muguruza and M. Tomas, *Deep learning volatility: a deep neural network perspective on pricing and calibration in (rough) volatility models*, Quantitative Finance, 21(1), 2021, pp. 11–27.

[14] A. Lewis, *A simple option formula for general jump-diffusion and other exponential Lévy processes*, working paper, 2001.

[15] R. Lord, R. Koekkoek and D. van Dijk, *A comparison of biased simulation schemes for stochastic volatility models*, Quantitative Finance, 10(2), 2010, pp. 177–194.

[16] R. Merton, *Theory of rational option pricing*, Bell Journal of Economics and Management Science, 4(1), 1973, pp. 141–183.

[17] M. Romano and N. Touzi, *Contingent claims and market completeness in a stochastic volatility model*, Mathematical Finance, 7(4), 1997, pp. 399–412.

[18] G. Willard, *Calculating prices and sensitivities for path-independent derivative securities in multifactor models*, Journal of Derivatives, 5(1), 1997, pp. 45–61.